

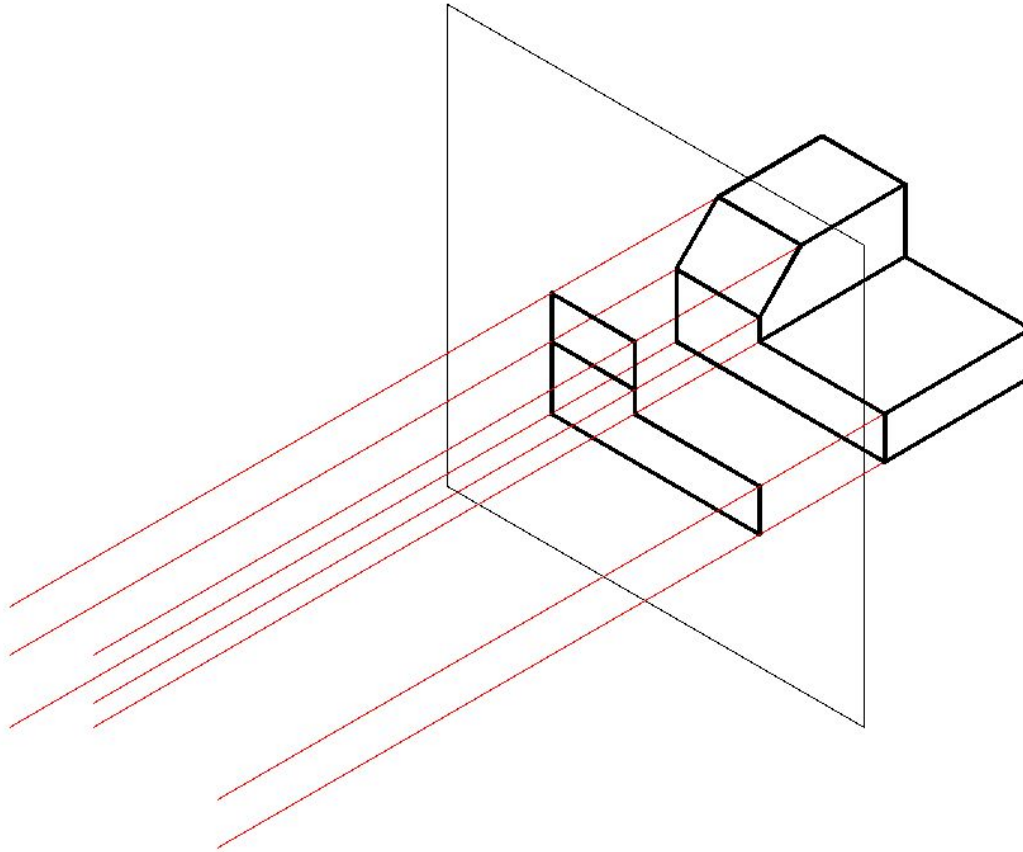
THEORY OF ORTHOGRAPHIC PROJECTION

1. PROJECTION

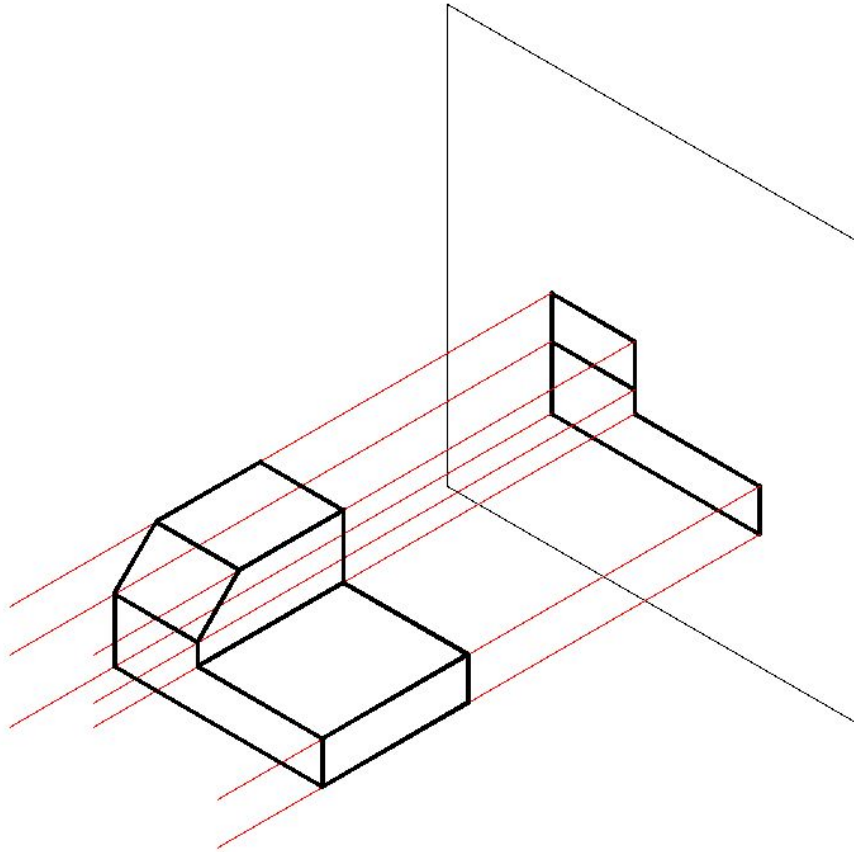
A method to describe shape by the process of causing an image to be formed by rays of sight taken in a particular direction from an object to a picture plane.

- **If the rays are perpendicular to the picture plane, the projection is known as *Orthographic Projection*.**
- **If the rays are at an angle to the plane, the projection is known as *Oblique Projection*.**
- **If the rays are taken to a particular station point, the projection is known as *Perspective Projection*.**

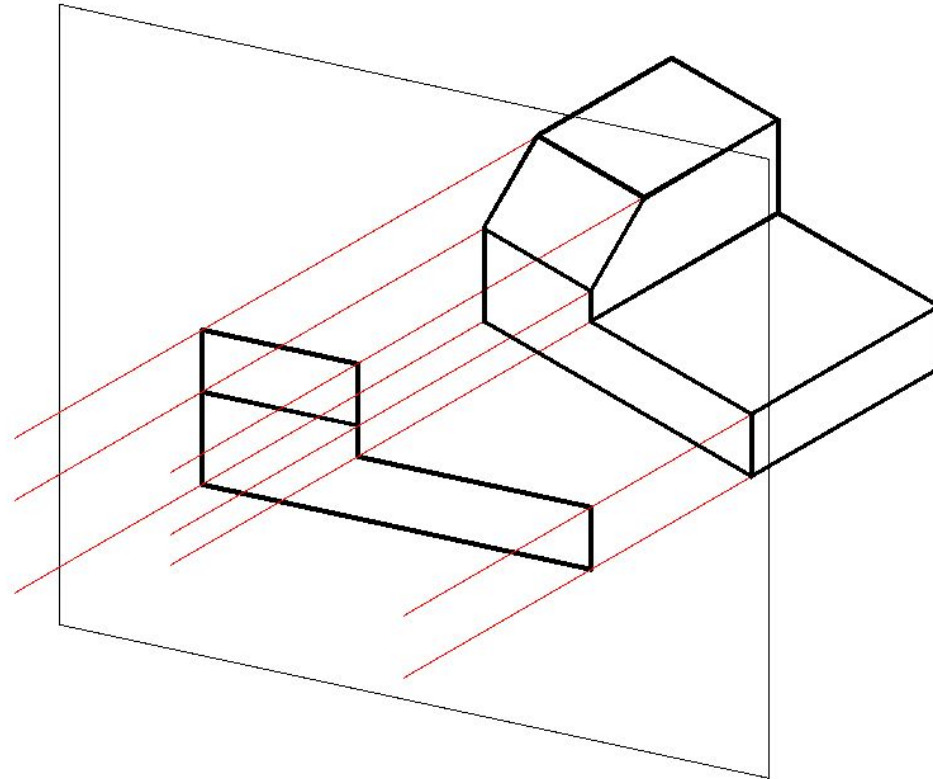
Orthographic Projection looking through the plane
(Plane between observer and object)



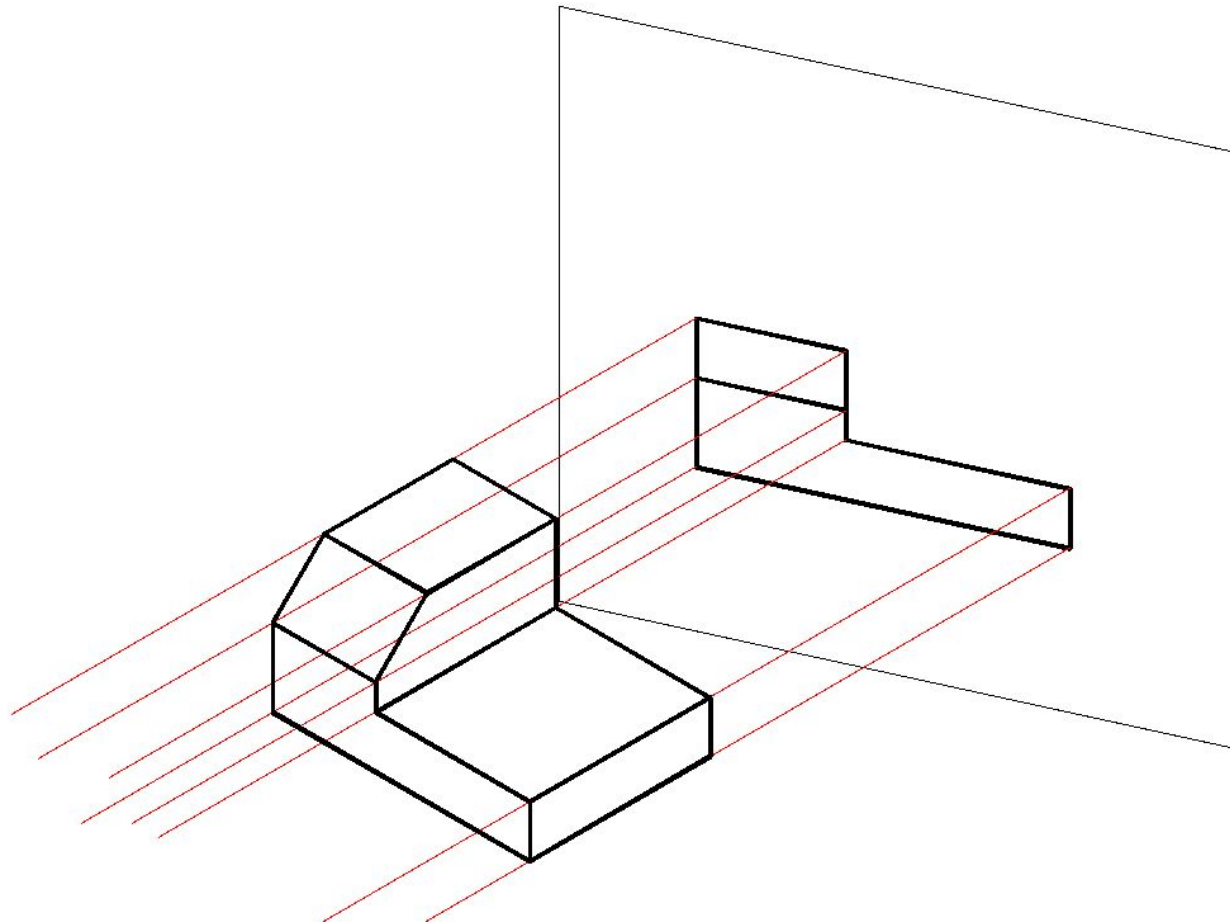
Orthographic Projection observing the object directly
(Object between Observer and Plane)



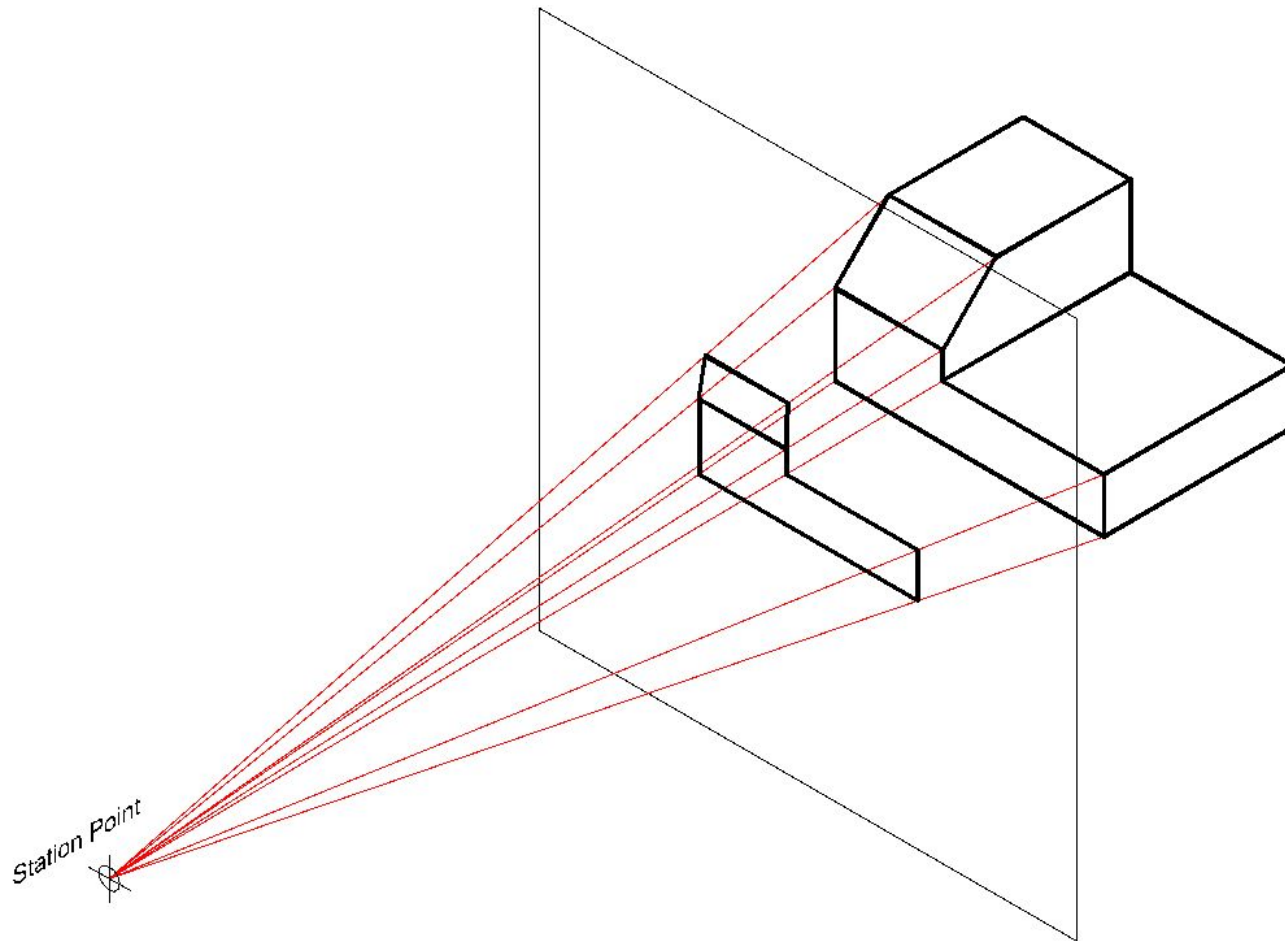
Oblique Projection Looking through the Plane (Plane between Observer and Object)



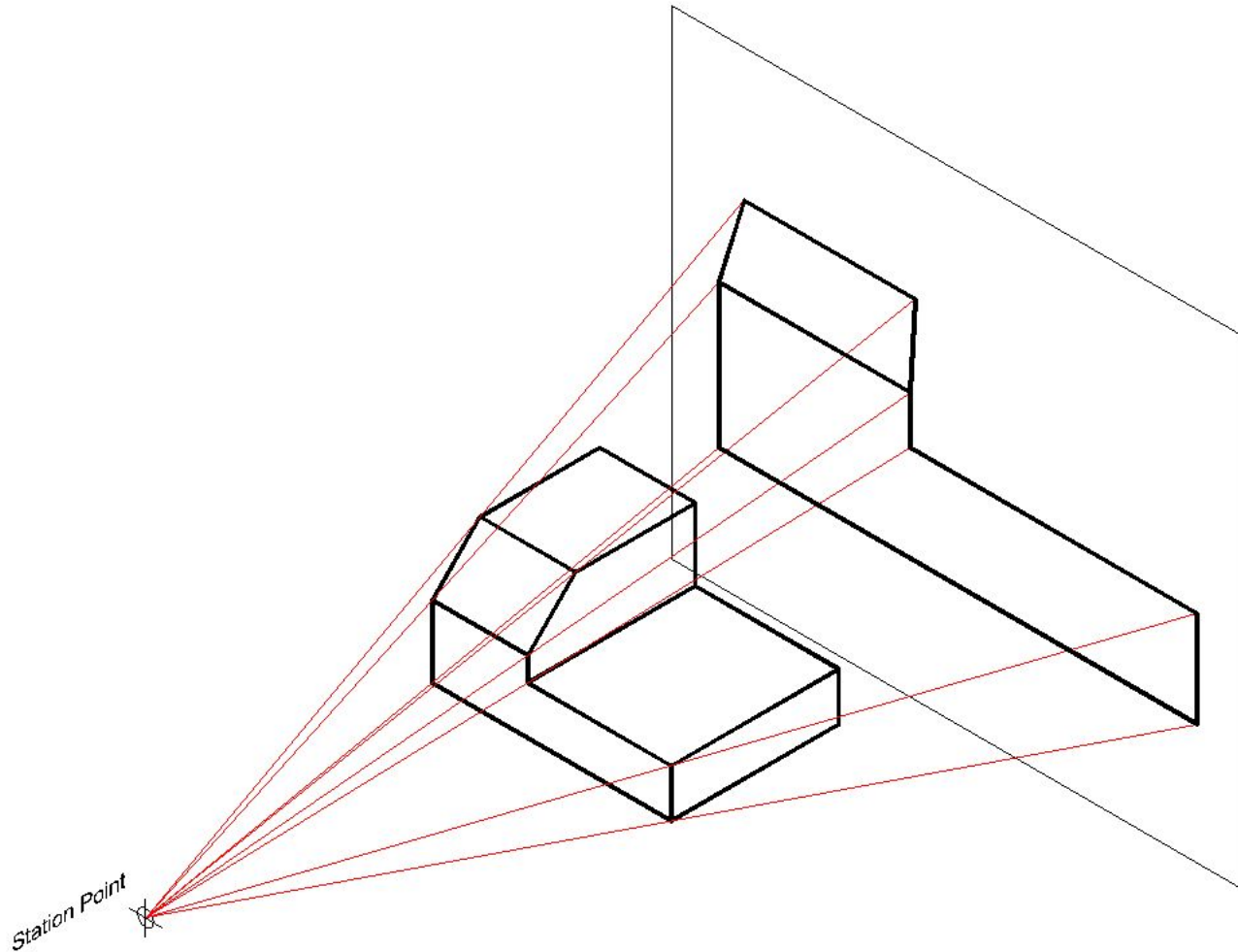
Orthographic Projection observing the object directly
(Object between Observer and Plane)



Perspective Projection looking through the plane (Plane between observer and object)



Perspective Projection observing the object directly
(Object between observer and Plane)



2. SHAPE REPRESENTATION METHODS

Two fundamental methods of shape representation are used:

a. Orthographic Views

These consists of a set of two or more separate views of an object taken from different directions, generally at right angles to each other and arranged relative to each other in a definite way. Each of the views shows the shape of the object for a particular view direction and collectively the views describe the object completely. Orthographic projection *only* is used.

b. Pictorial Views

In this method, the object is oriented either behind or in front of a single plane and projected upon it. Either orthographic, oblique, or perspective projection is used.

3. PRINCIPAL PLANES OF PROJECTION

a. Vertical Plane

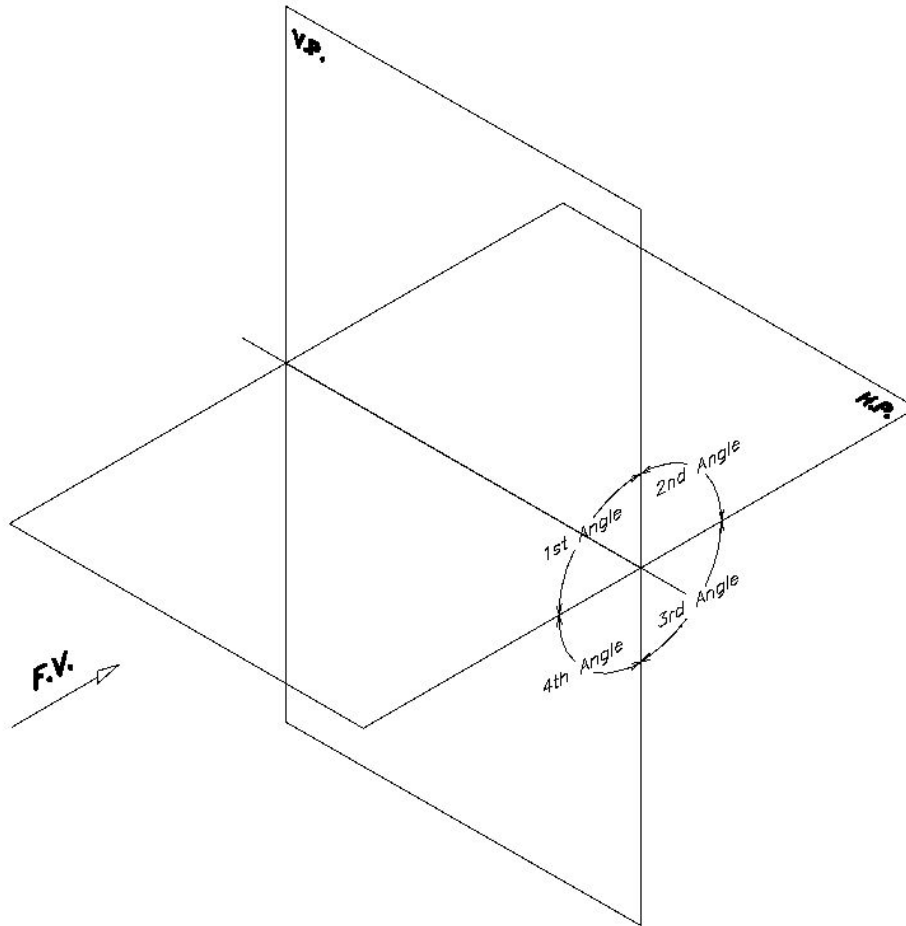
Used for the projection of Front View.

b. Horizontal Plane

Used for the projection of Top View.

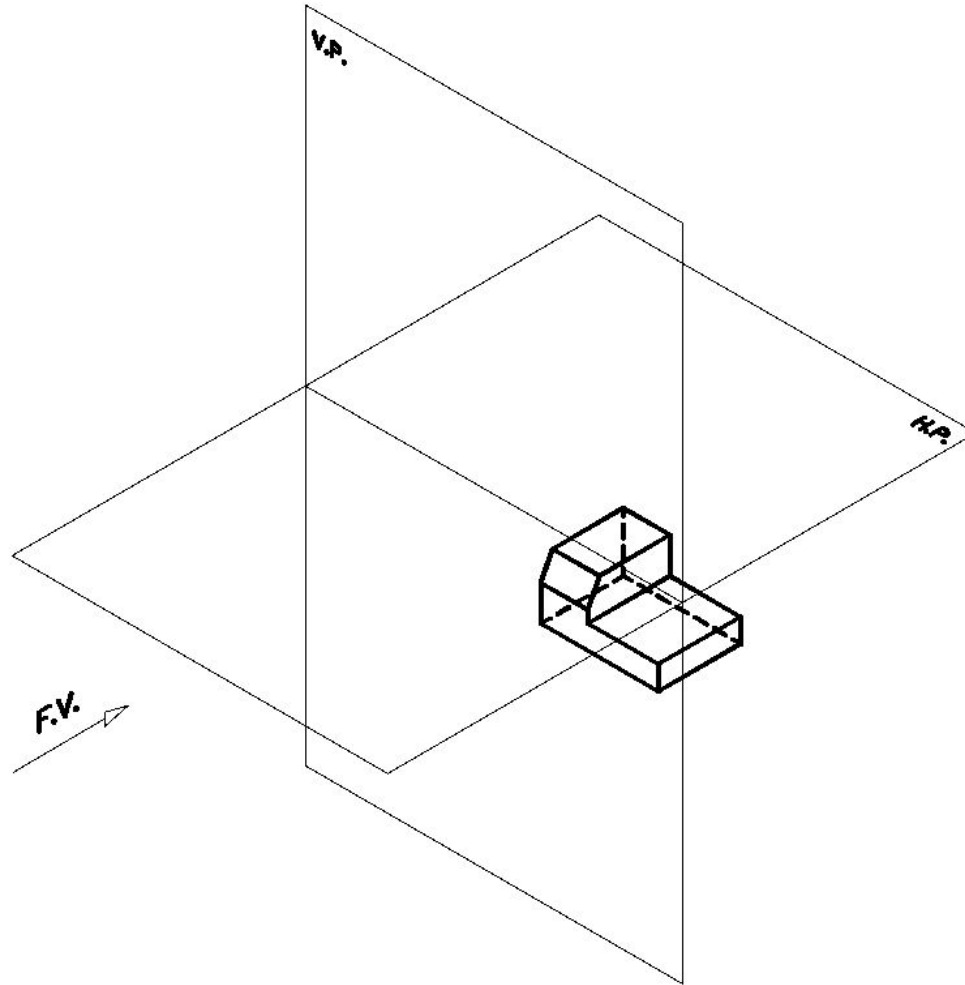
The horizontal plane and the vertical plane intersect at right angle and create a line of intersection known as the reference line. This intersection also creates four quadrants.

PRINCIPAL PLANES OF PROJECTION



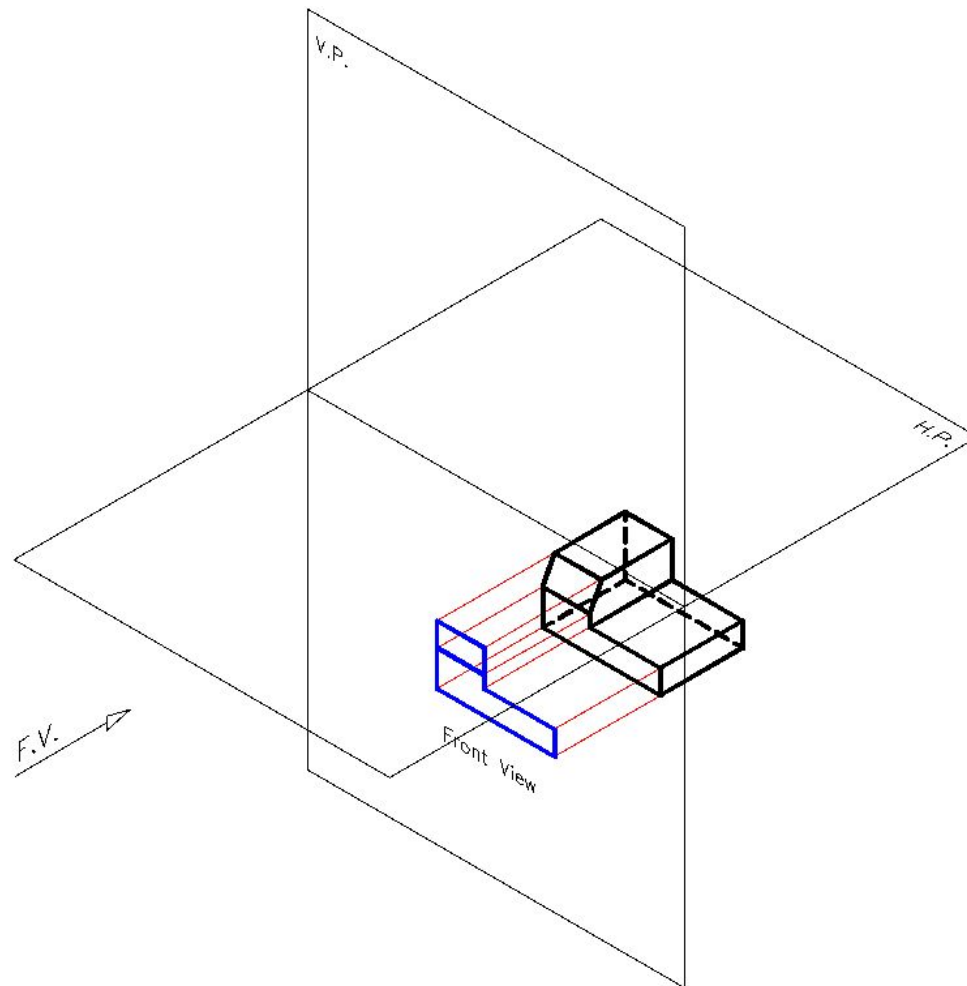
4. THIRD ANGLE PROJECTION METHOD

Plane of projection is always between the object and observer.



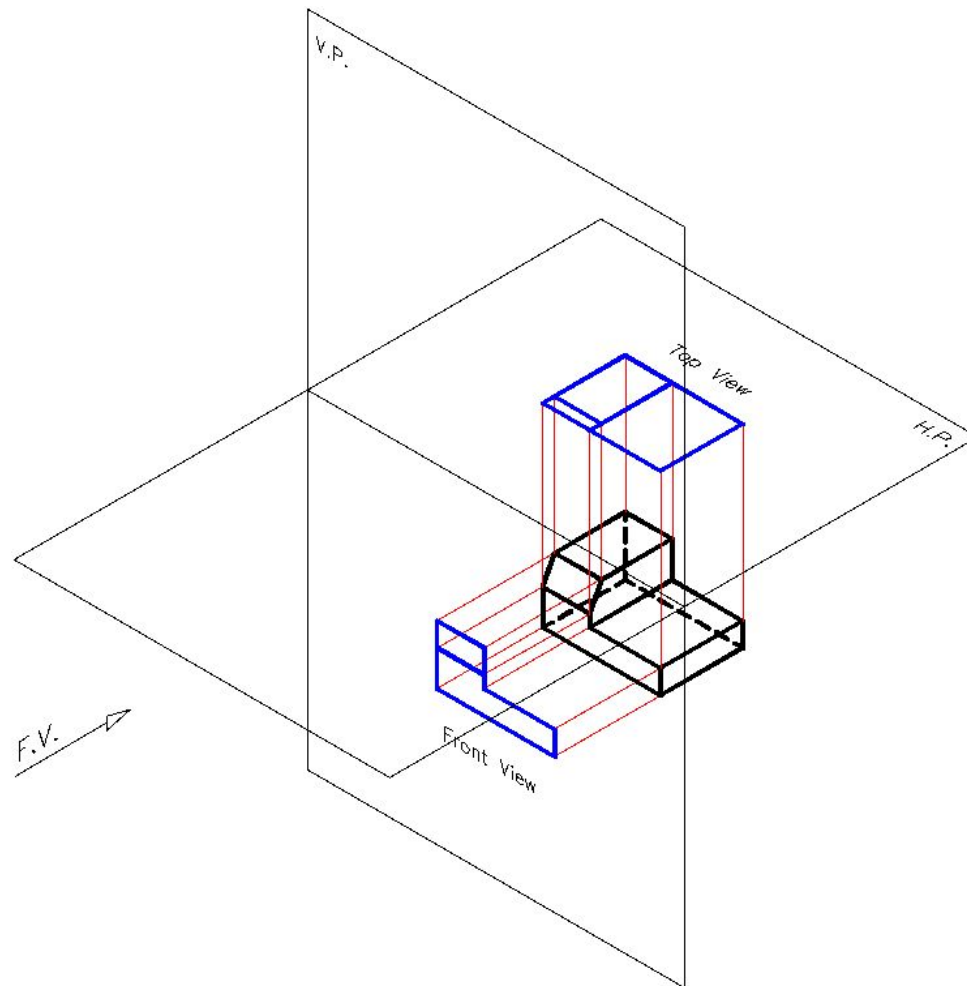
THIRD ANGLE PROJECTION METHOD

Projection of Front View



THIRD ANGLE PROJECTION METHOD

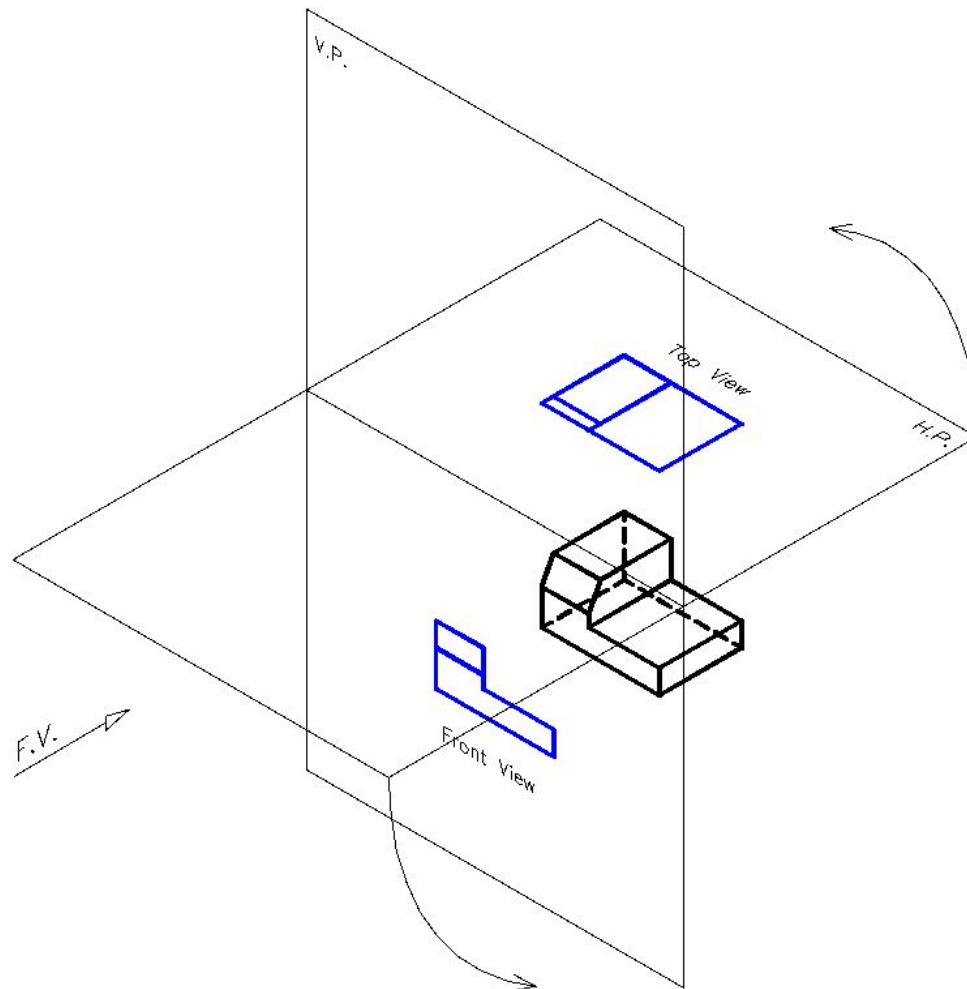
Projection of Top View



THIRD ANGLE PROJECTION METHOD

Bringing Front View and Top View in One Plane

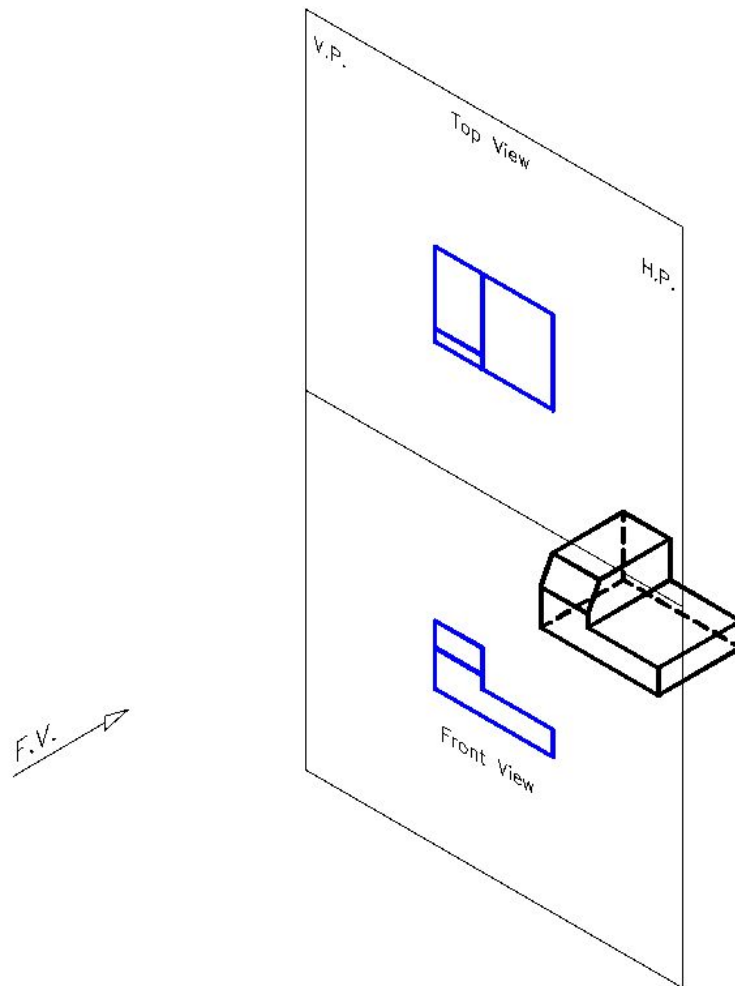
(Always open 1st Angle by rotating the H.P. about the V.P.)



THIRD ANGLE PROJECTION METHOD

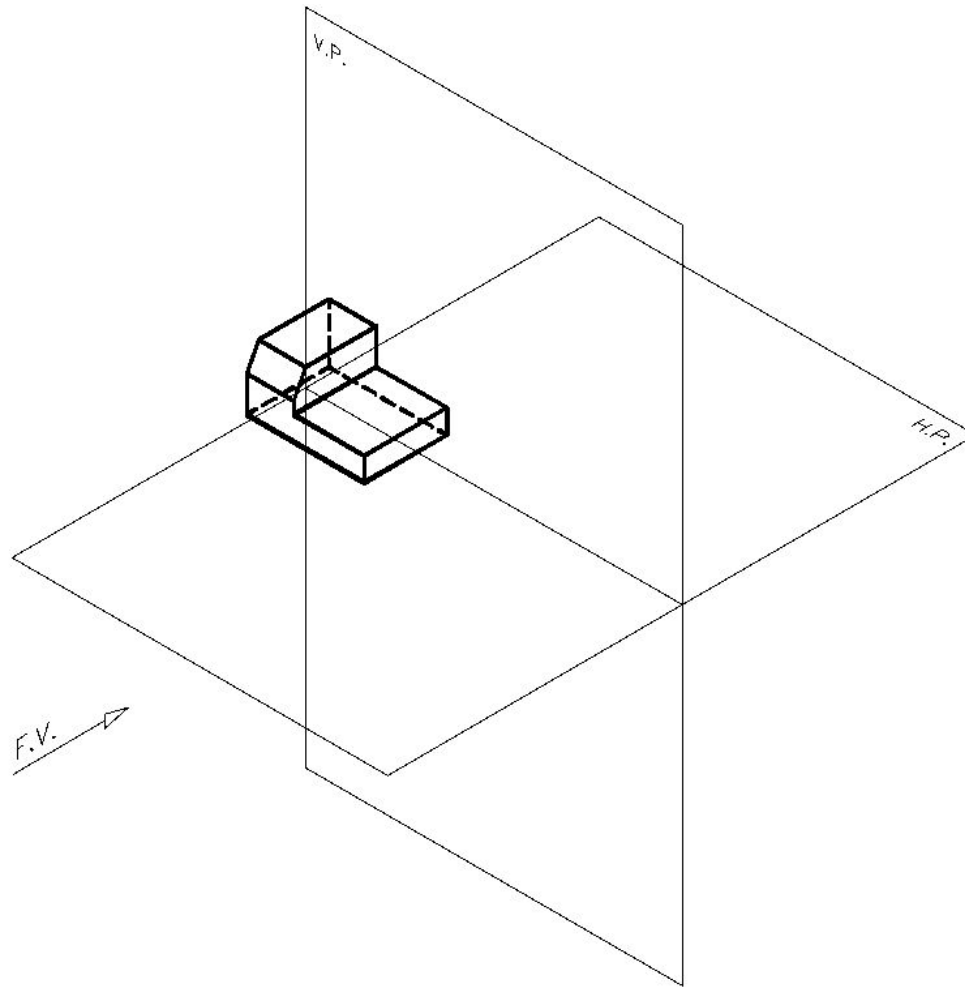
Bringing Front View and Top View in One Plane

(Always open 1st Angle by rotating the H.P. about the V.P.)



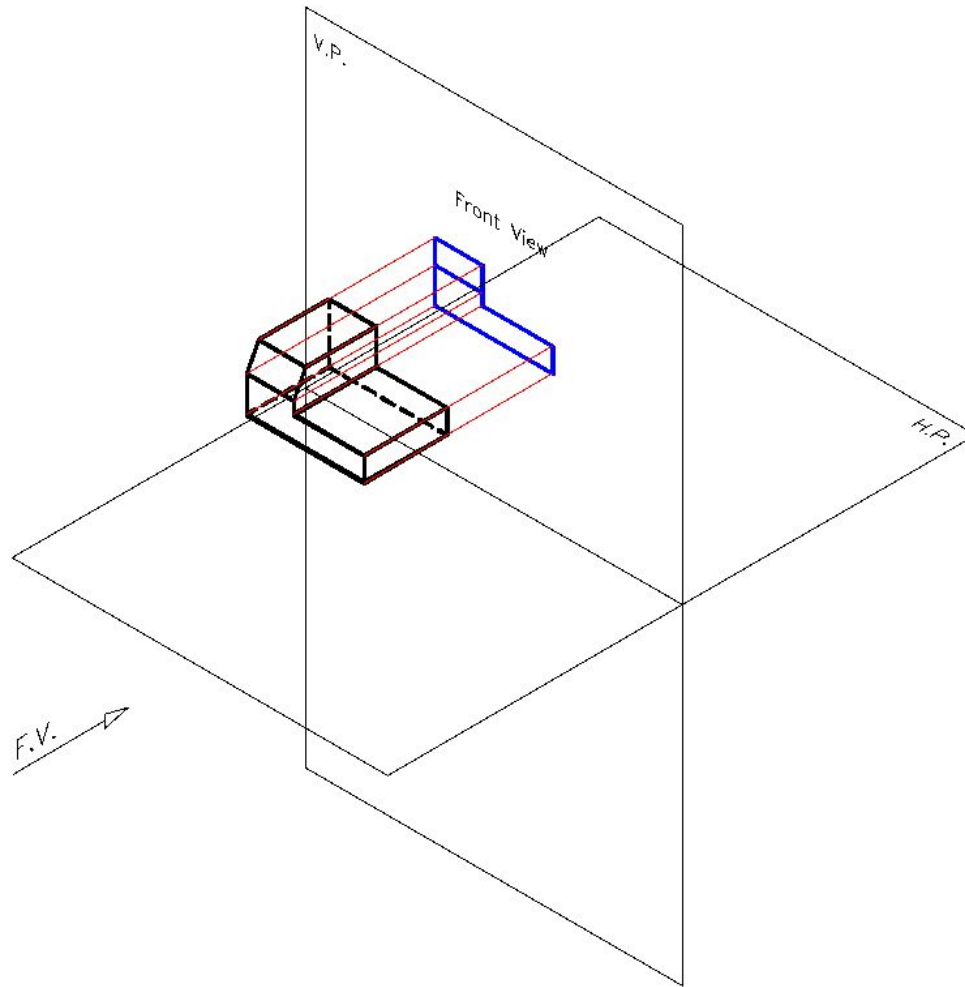
5. FIRST ANGLE PROJECTION METHOD

Object is always between the plane of projection and the observer.



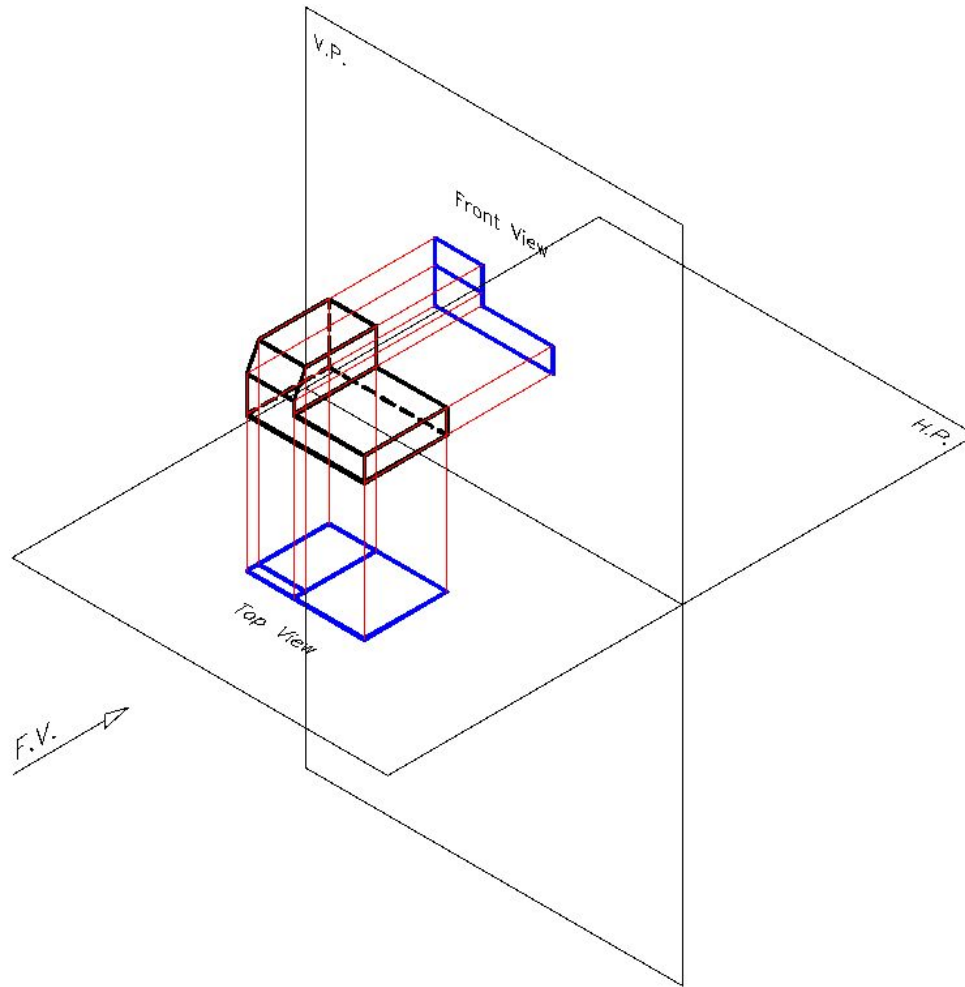
FIRST ANGLE PROJECTION METHOD

Projection of Front View



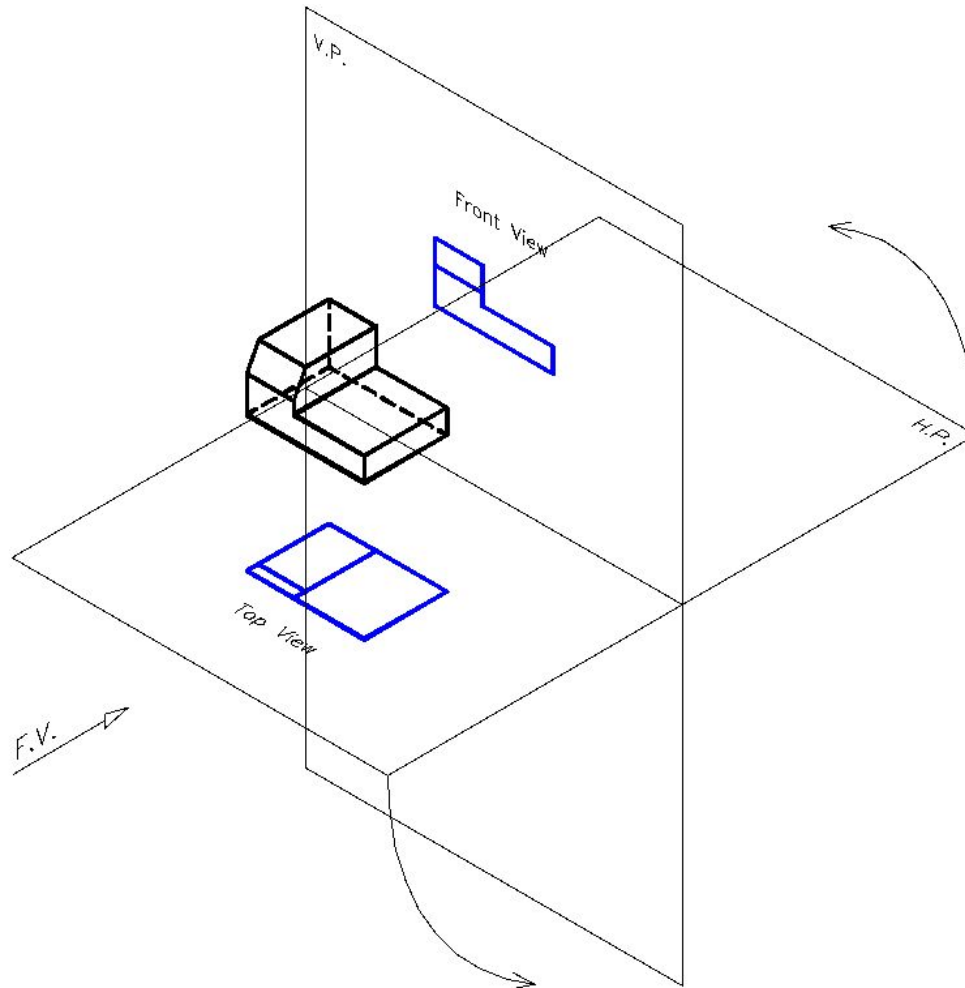
FIRST ANGLE PROJECTION METHOD

Projection of Top View



FIRST ANGLE PROJECTION METHOD

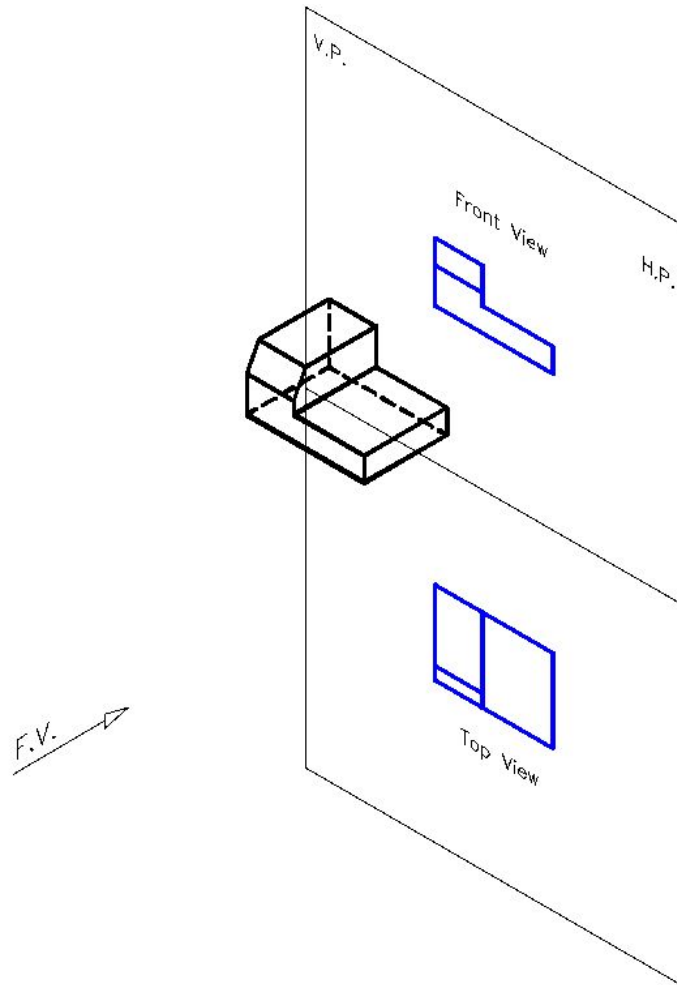
Bringing Front View and Top View in One Plane
(Always open 1st Angle by rotating the H.P. about the V.P.)



FIRST ANGLE PROJECTION METHOD

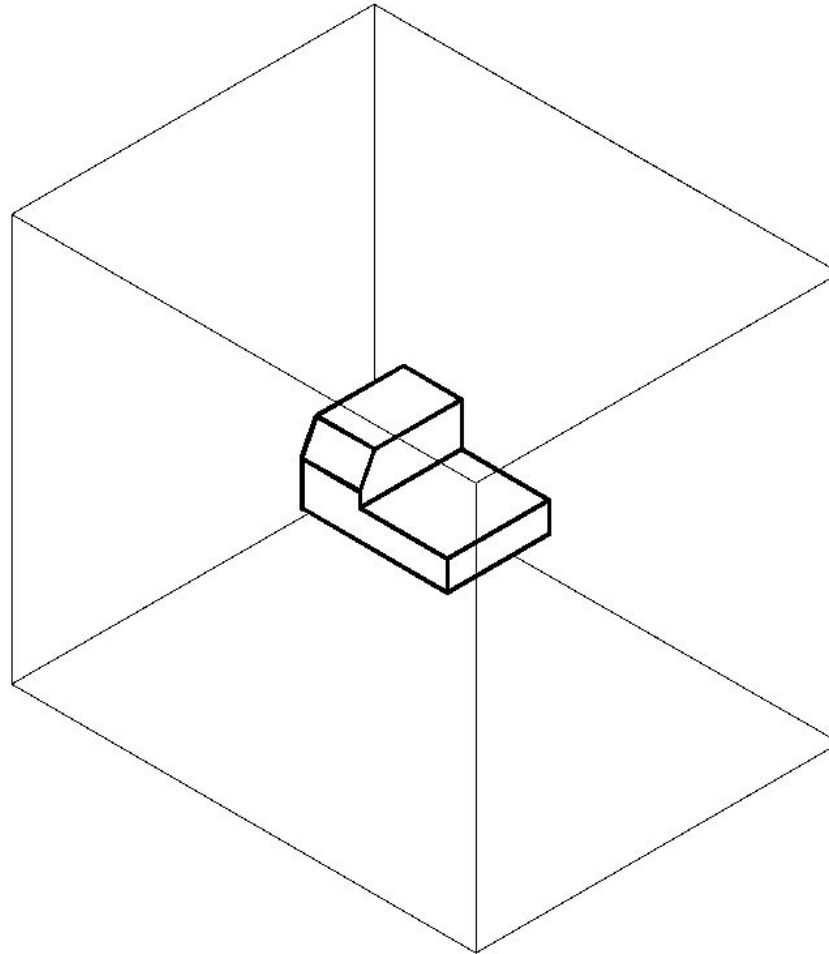
Bringing Front View and Top View in One Plane

(Always open 1st Angle by rotating the H.P. about the V.P.)

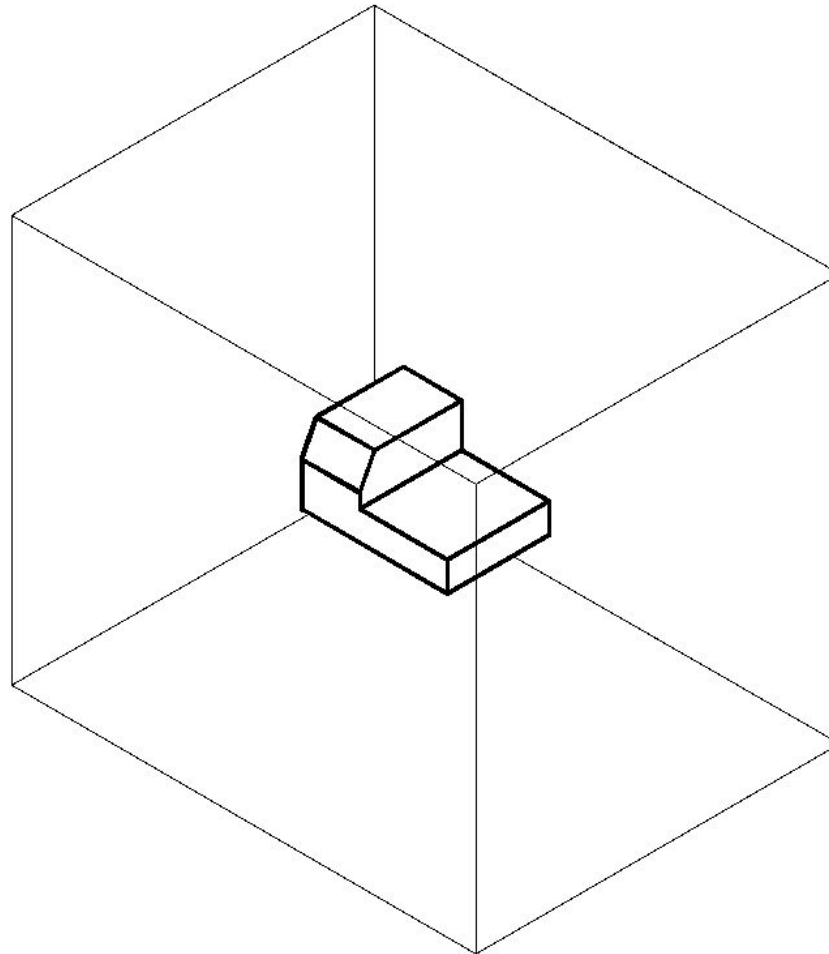


6. THE SIX PRINCIPAL VIEWS

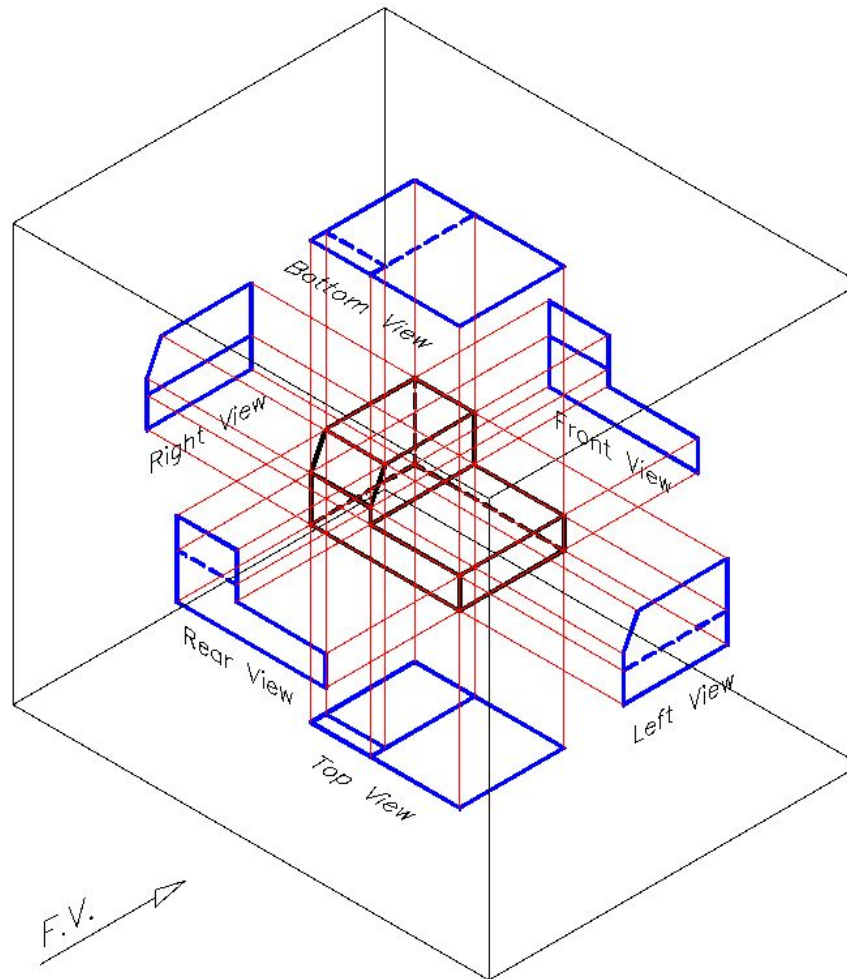
The object can be entirely surrounded by a set of six transparent planes, each at right angles to the four adjacent to it.



- **The object is in a transparent box.**
- **In the First Angle Projection method, the observer stands inside the box, thus keeping the object between him and the plane of projection.**
- **In the Third Angle Projection method, the observer stands outside the box, keeping the plane of projection always between him and the object.**

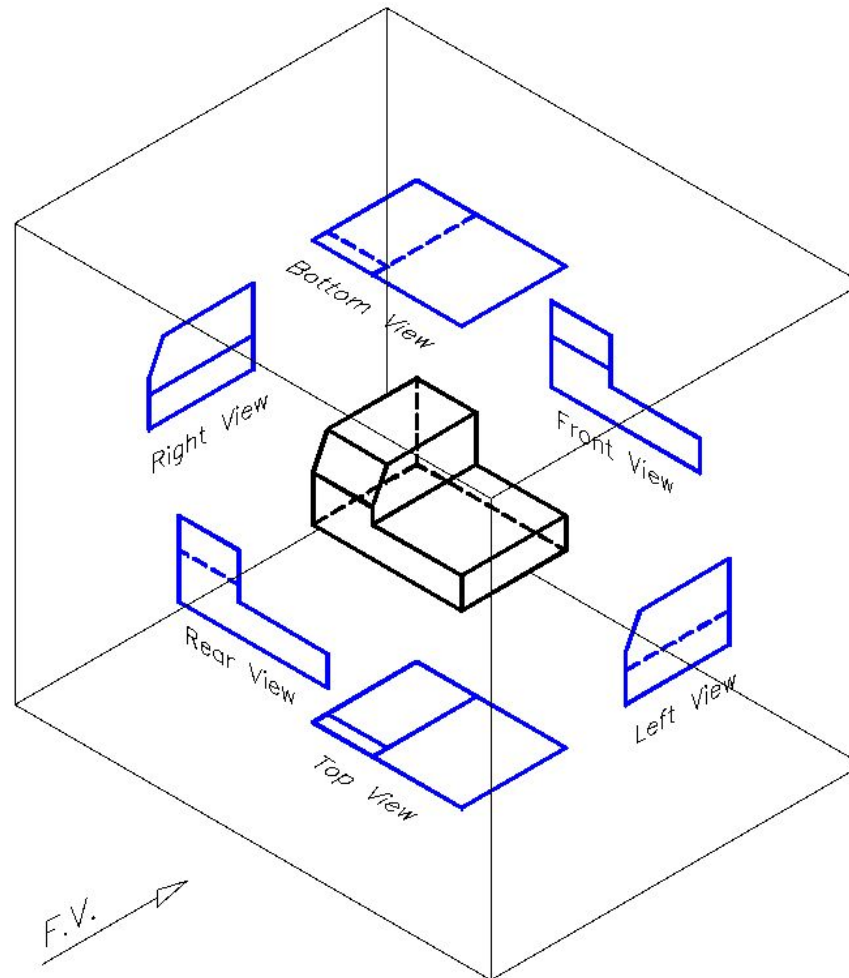


- In the First Angle Projection method, the views are projected to the other side of the object.

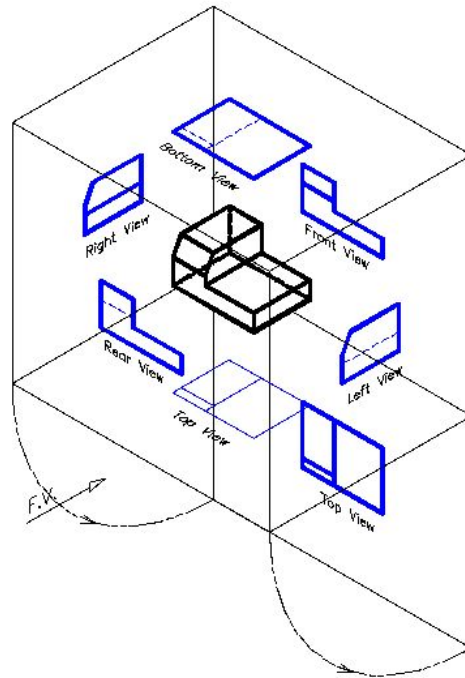


Placement of Six Views in the First Angle Projection Method

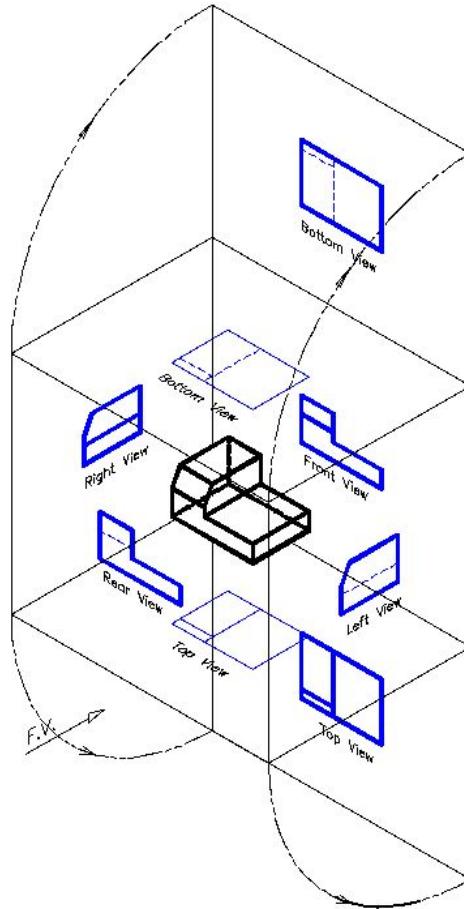
(Observe the object from inside the box)



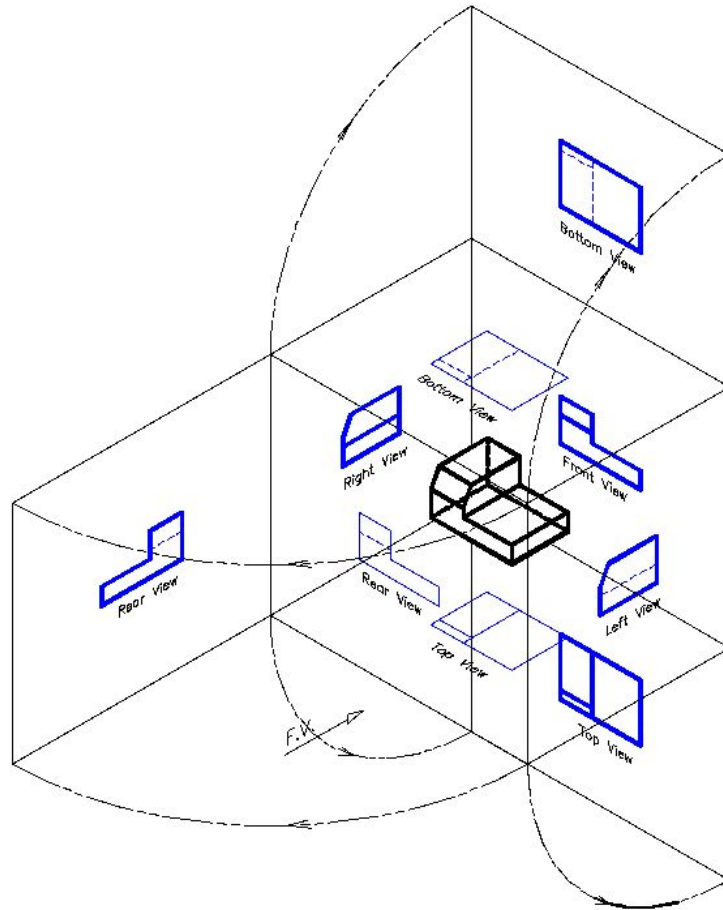
Opening the Transparent Box to bring the six views in one plane (1st Angle)



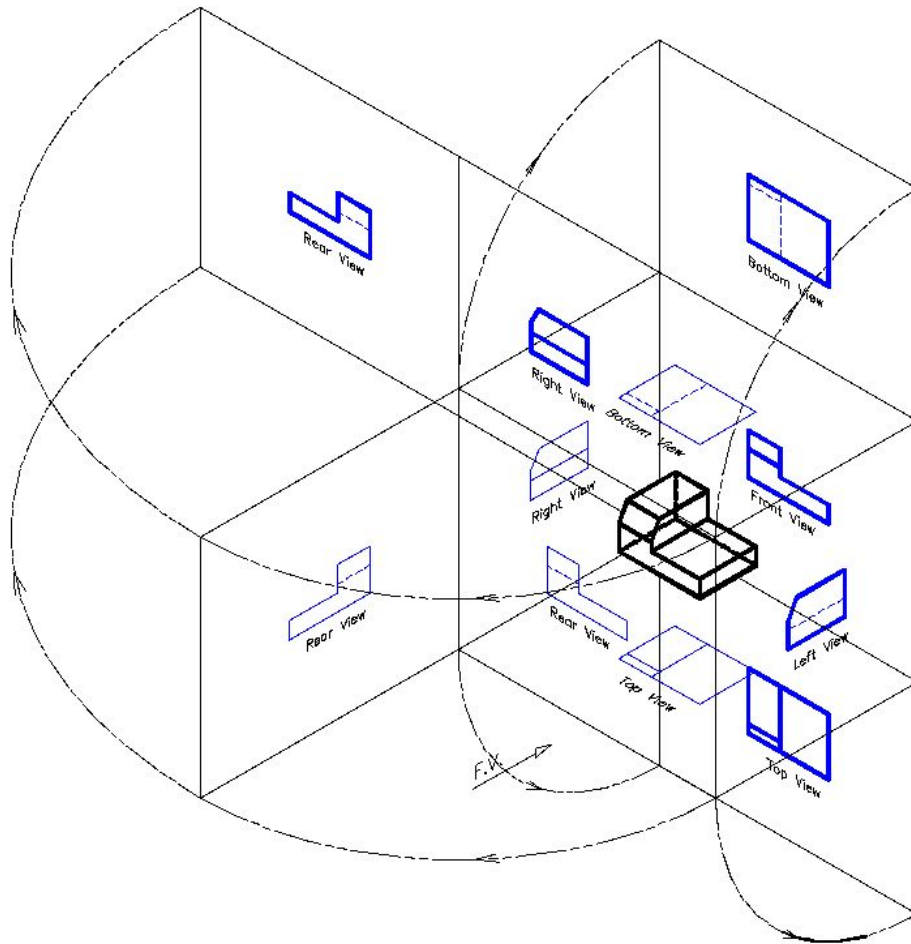
Opening the Transparent Box to bring the six views in one plane (1st Angle)



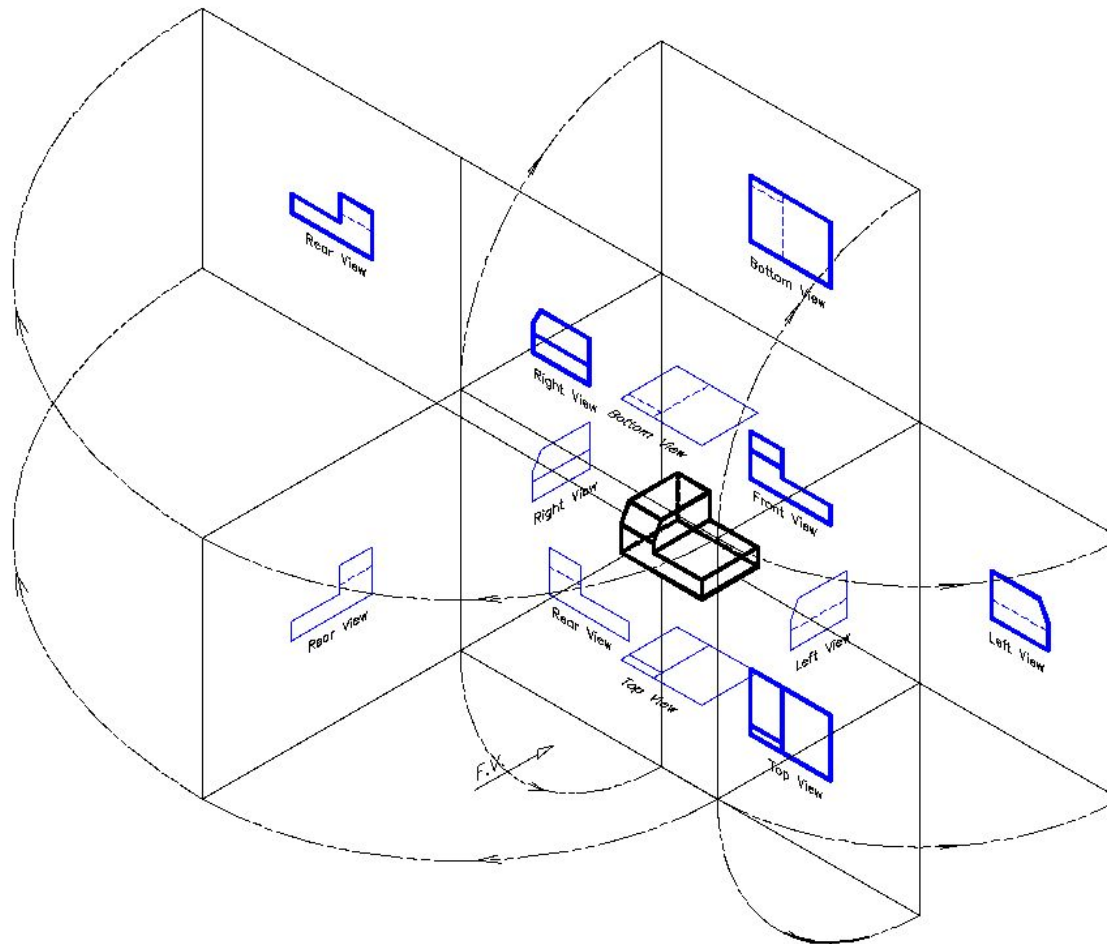
Opening the Transparent Box to bring the six views in one plane (1st Angle)



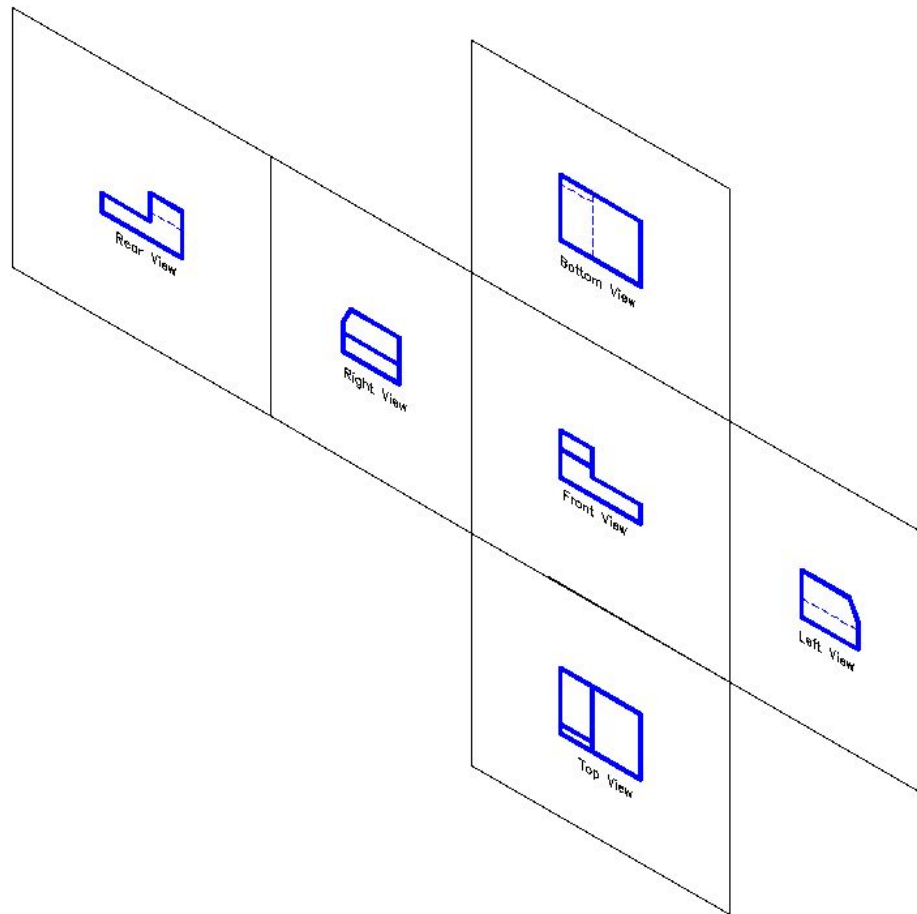
Opening the Transparent Box to bring the six views in one plane (1st Angle)



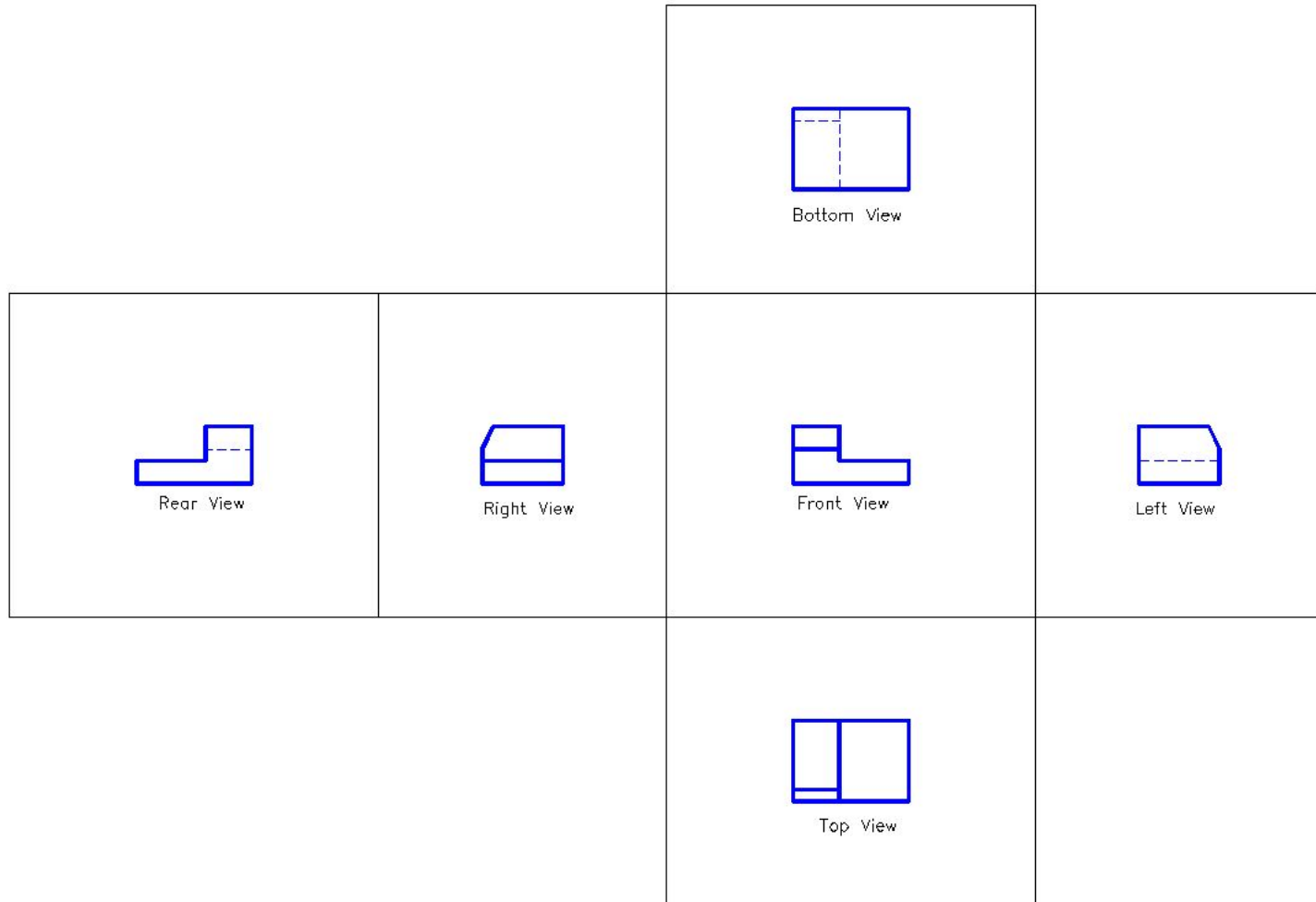
Opening the Transparent Box to bring the six views in one plane (1st Angle)



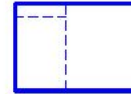
The six views in one plane (1st Angle)



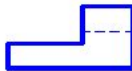
The six views in one plane (1st Angle)



The six views in one plane (1st Angle)



Bottom View



Rear View



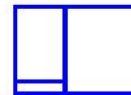
Right View



Front View

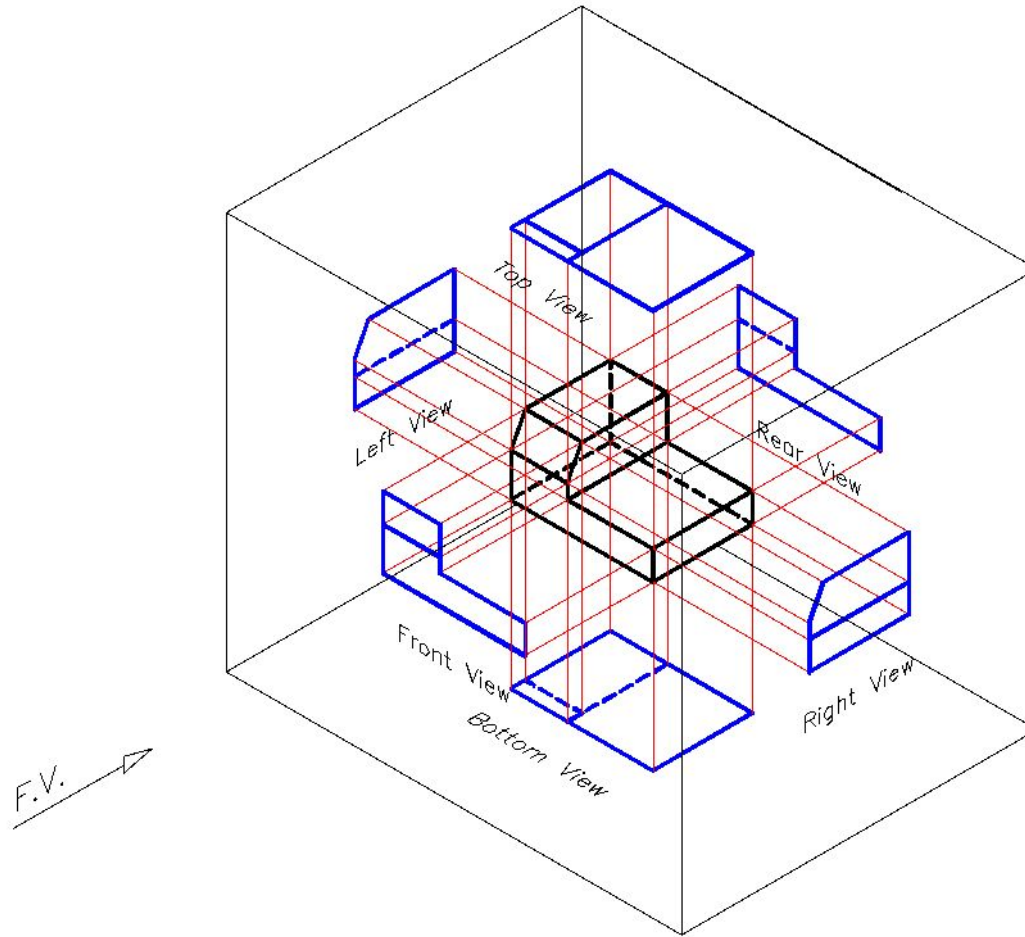


Left View

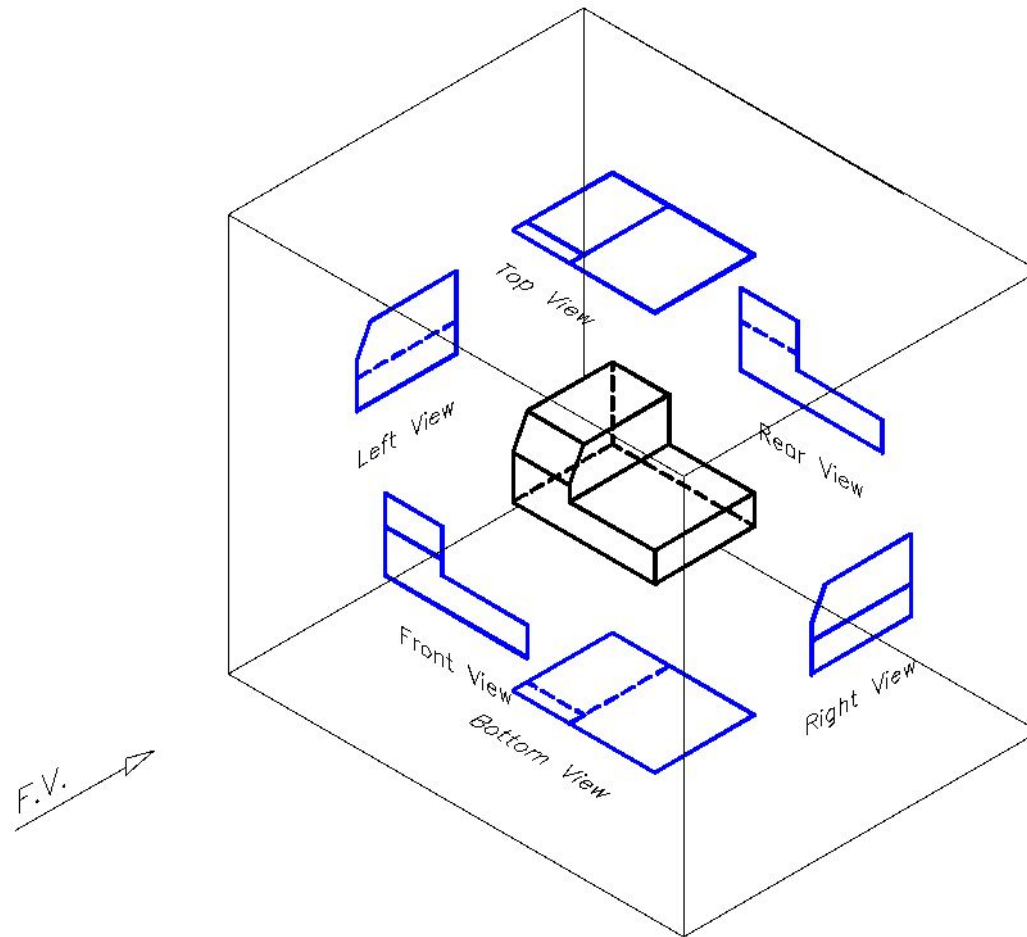


Top View

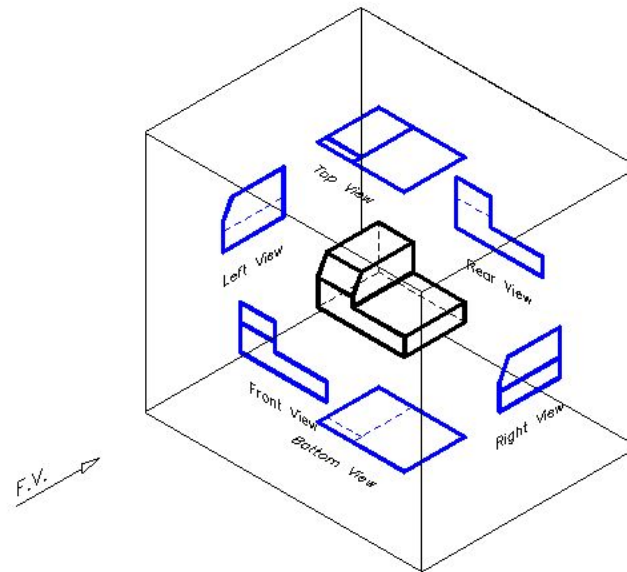
- In the Third Angle Projection method, the views are projected to the same side of the object.



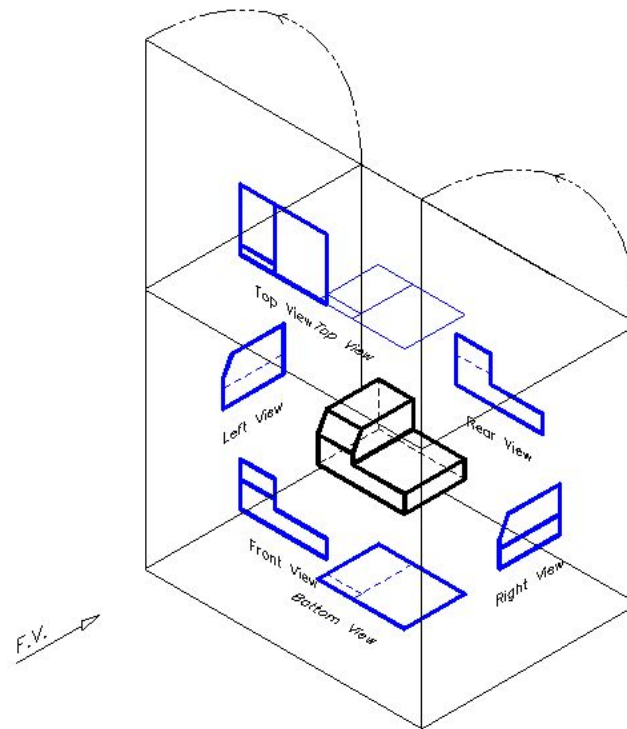
Placement of Six Views in the Third Angle Projection Method (Observe the object from outside the box)



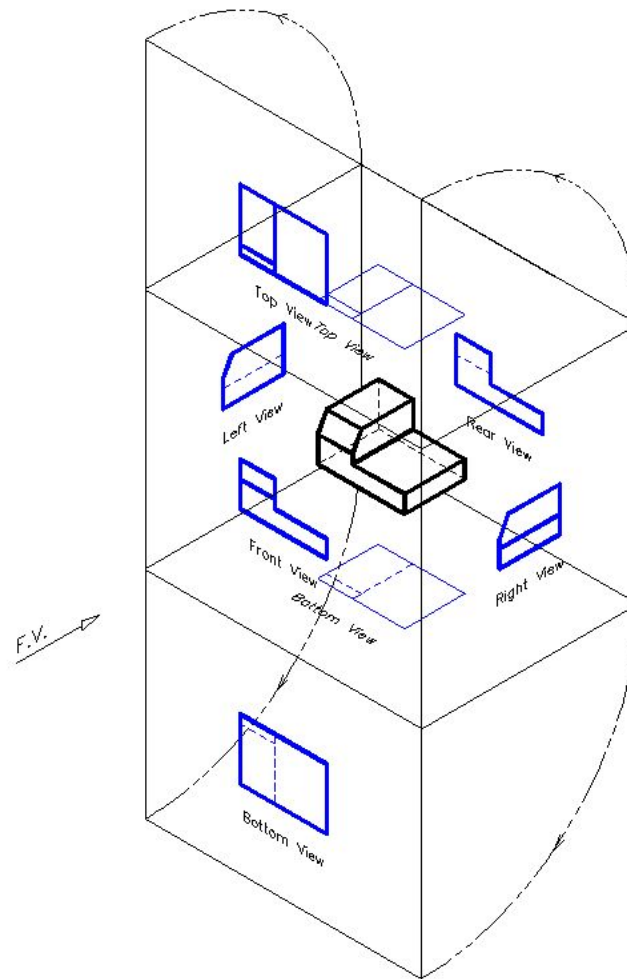
Opening the Transparent Box to bring the six views in one plane (3rd Angle)



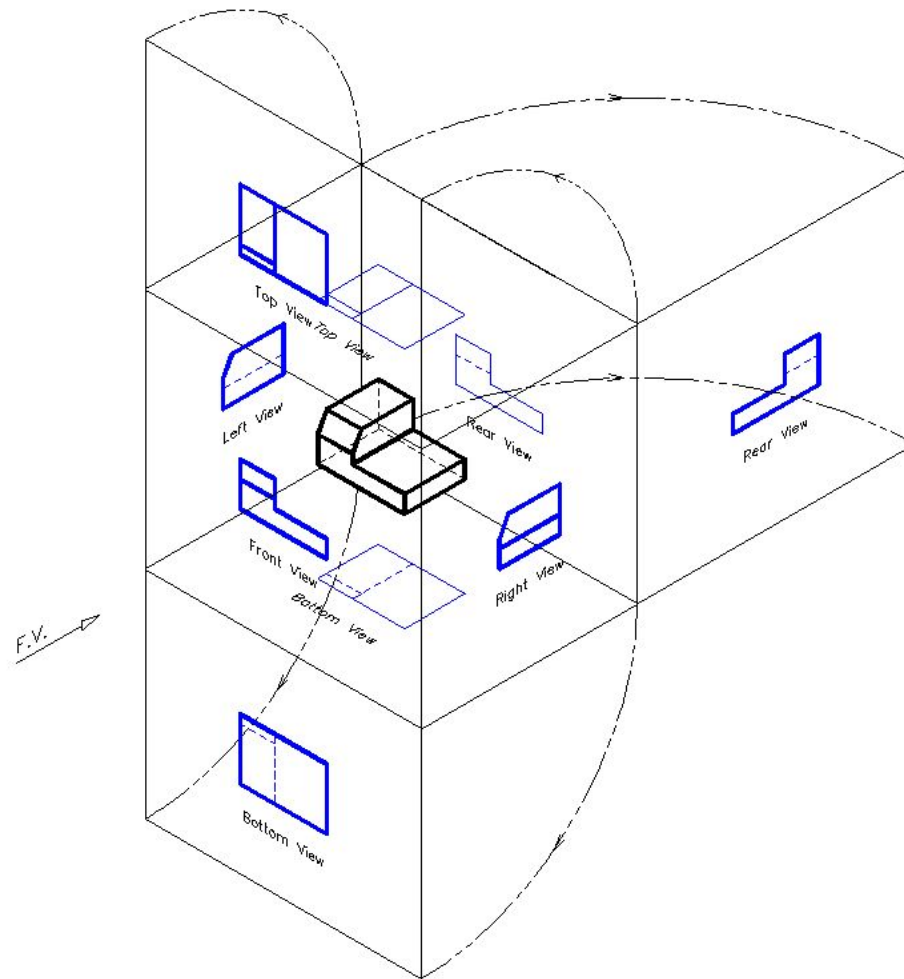
Opening the Transparent Box to bring the six views in one plane (3rd Angle)



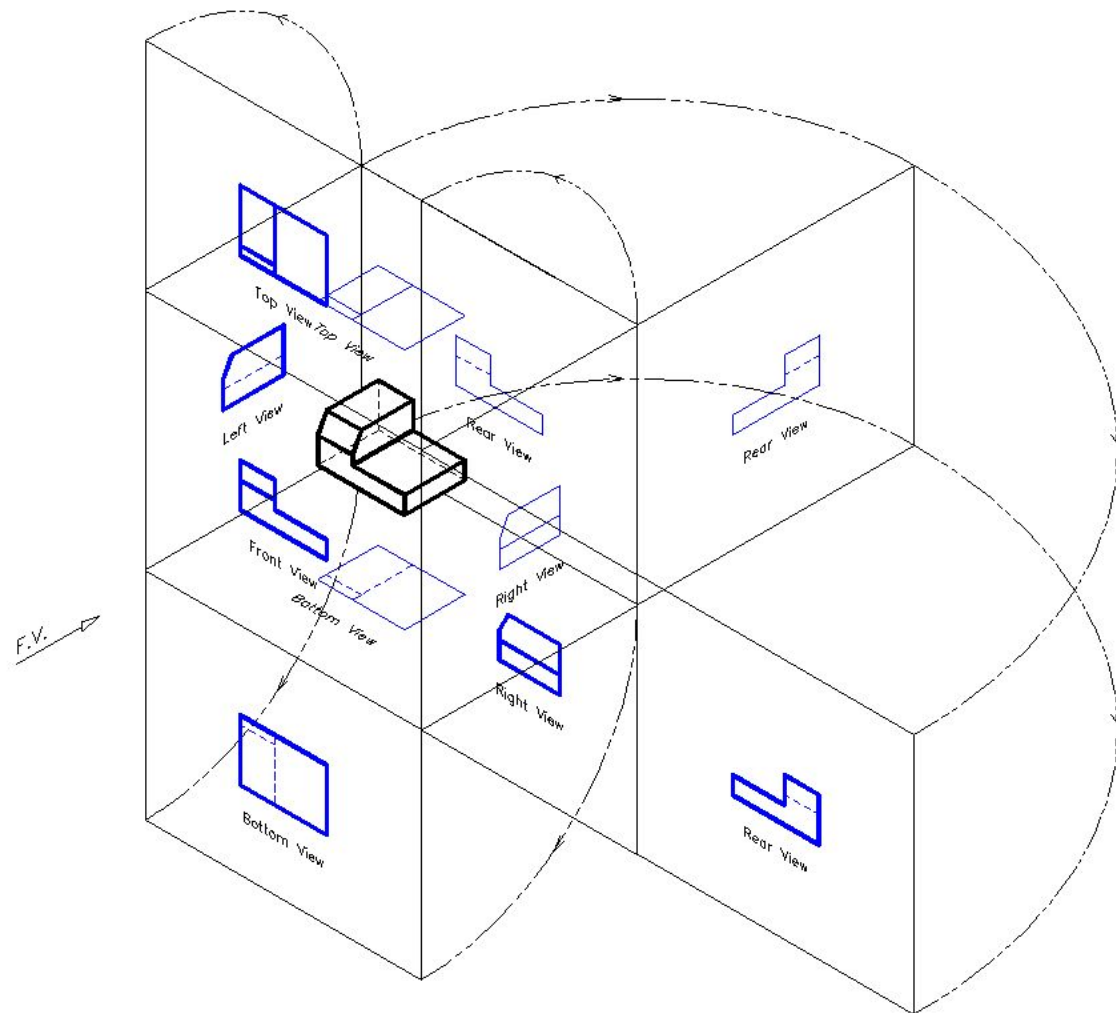
Opening the Transparent Box to bring the six views in one plane (3rd Angle)



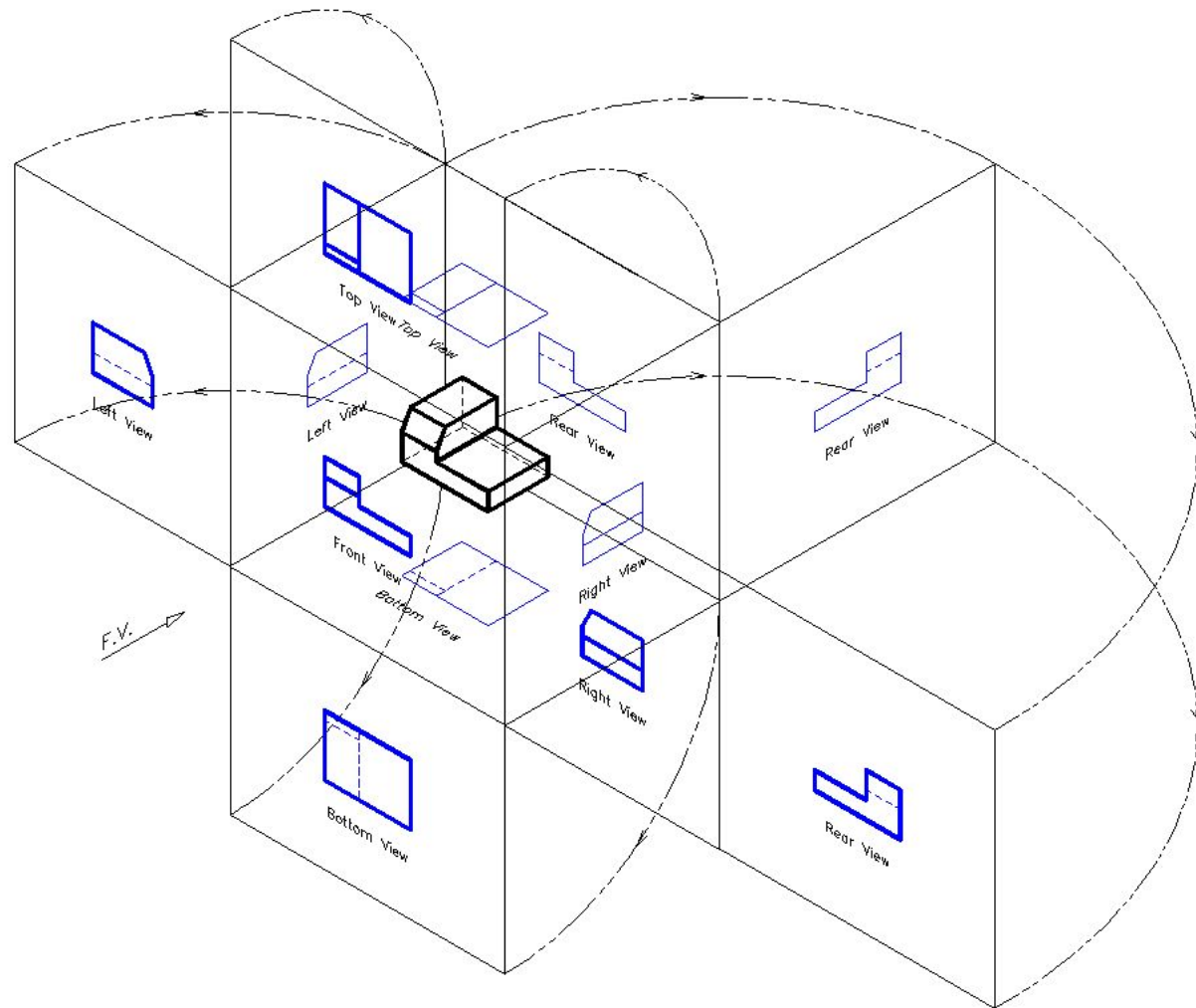
Opening the Transparent Box to bring the six views in one plane (3rd Angle)



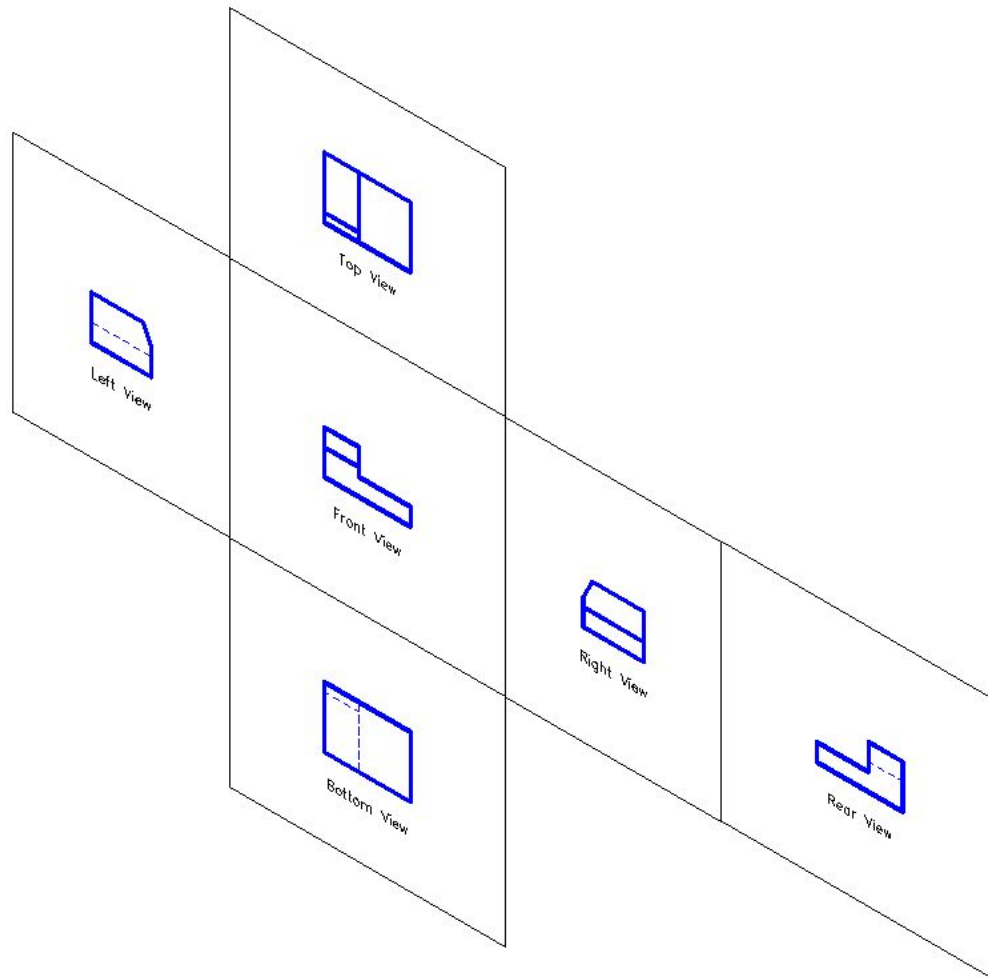
Opening the Transparent Box to bring the six views in one plane (3rd Angle)



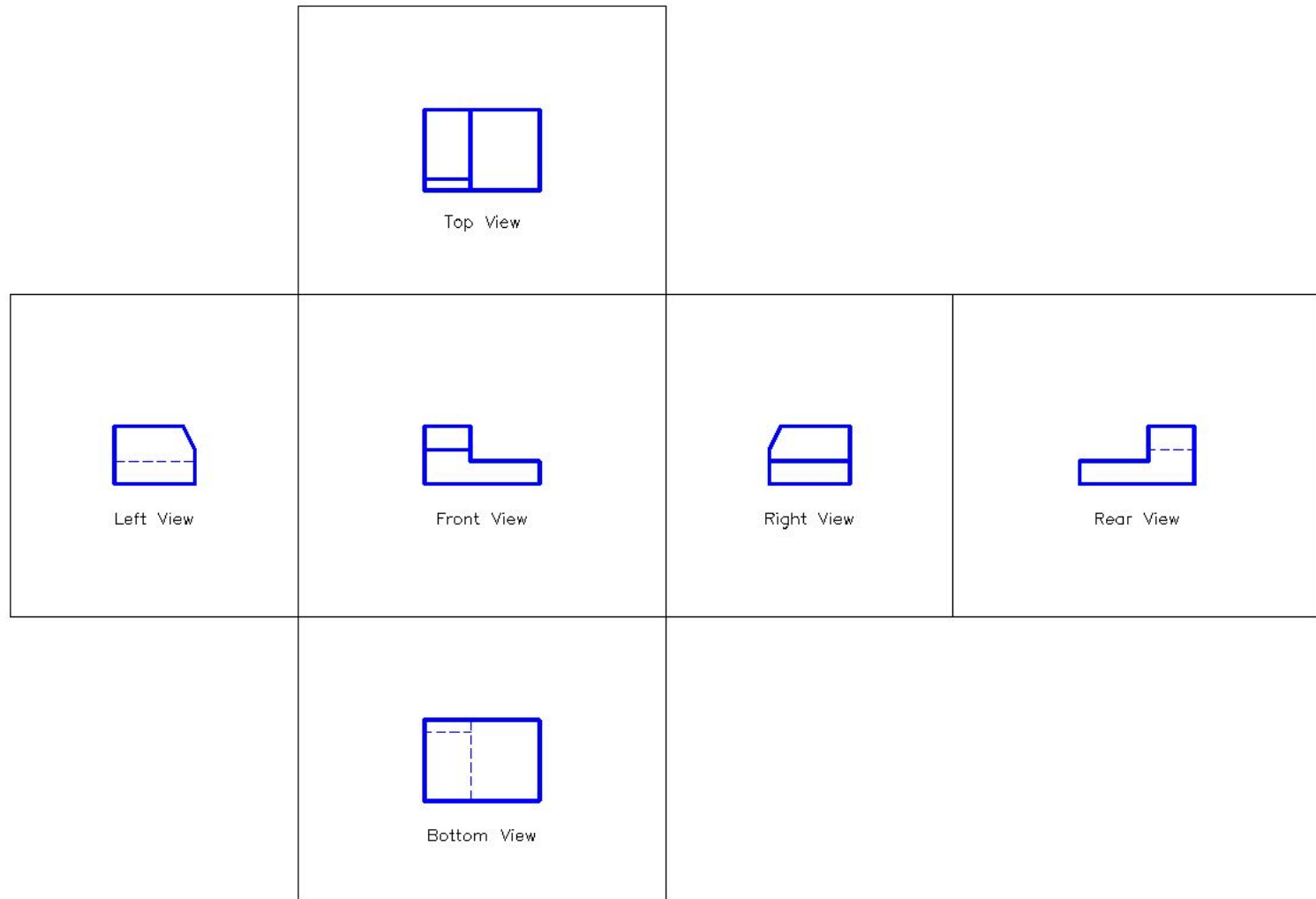
Opening the Transparent Box to bring the six views in one plane (3rd Angle)



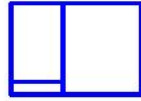
The Six Views in One Plane (3rd Angle)



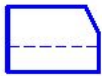
The Six Views in One Plane (3rd Angle)



The Six Views in One Plane (3rd Angle)



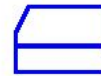
Top View



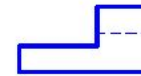
Left View



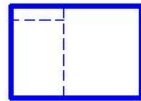
Front View



Right View



Rear View



Bottom View

USING ORTHOGRAPHIC PROJECTIONS

7. OBJECT ORIENTATION

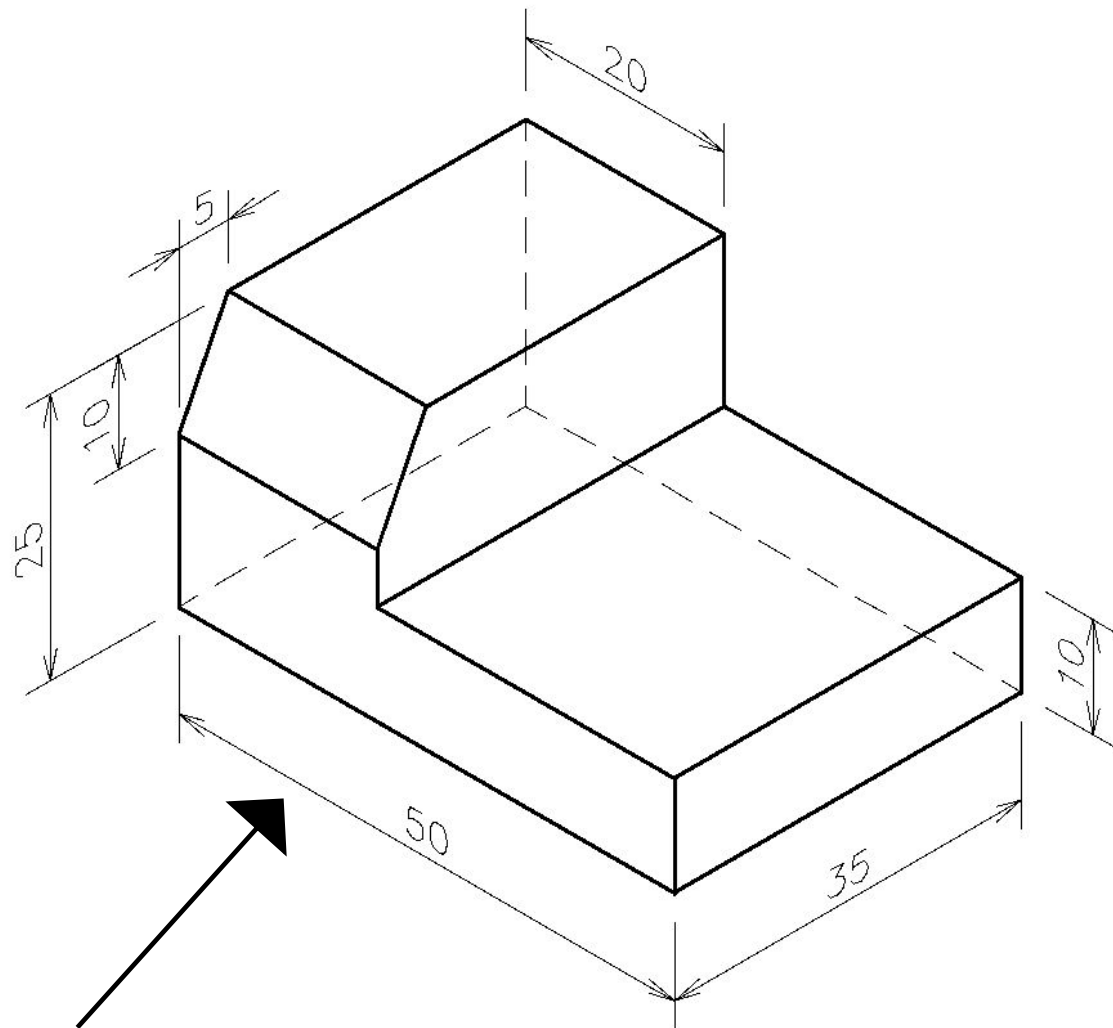
An object can be drawn in several possible positions. In general the simplest position shall be used. The object shall be positioned in such a way so that the principal faces shall lie perpendicular to the direction of vision.

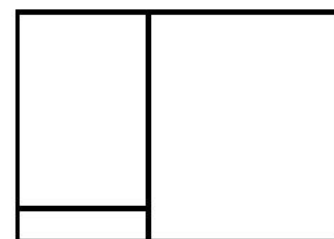
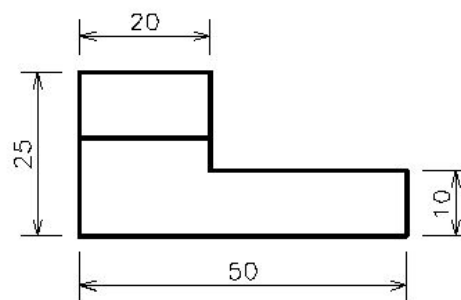
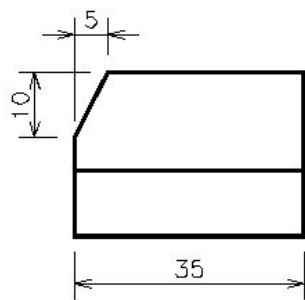
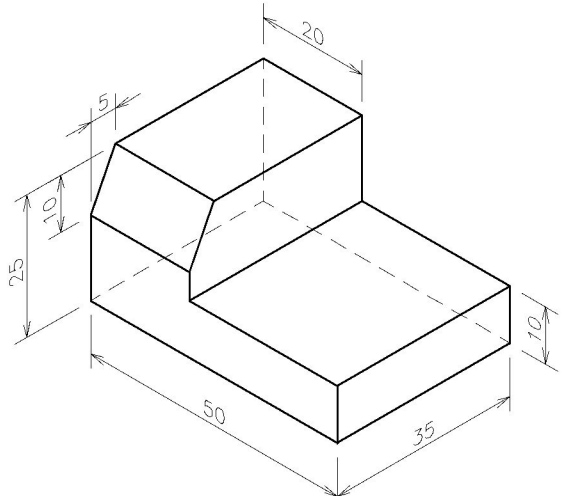
8. SELECTION OF VIEWS

It is important to choose the combination of views that will describe the shape of the object in the best and most economical way. Often only two views are necessary. However, in some case three or more views might be required to fully describe the object. Unnecessary views shall be avoided.

9. VIEW SPACING

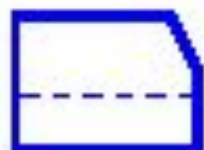
View spacing is necessary so that the drawing will be balanced within the space provided.







Front View



Left View



Top View